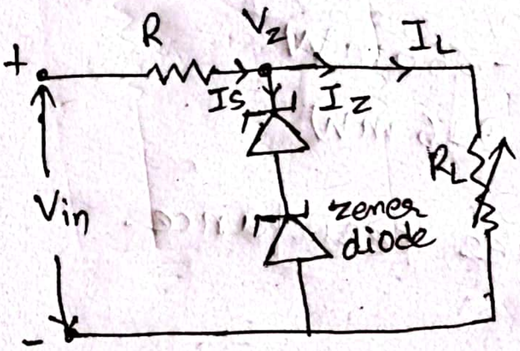
Diode (Prabin Chaudhary)

2021- Spring

1. A zener ~~diode~~ regulator consists of two zener diode with 6V zener voltage each, with variable load resistance from 200 Ω to 1K Ω. Given that below Vin = 15V and R = 100 Ω.

Sol: The given zener regulator is:



$R_{L(min)} = 200\,\Omega$

$R_{L(max)} = 1K\Omega = 1000\,\Omega$

$V_{in} = 15\,V$

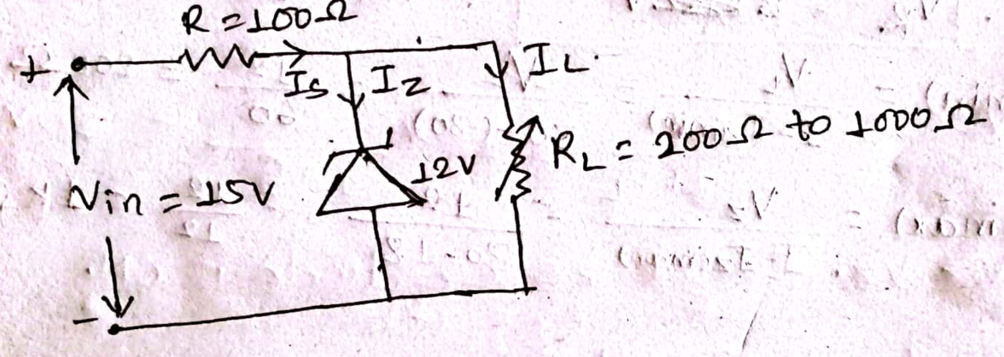
$R = 100\,\Omega$

$V_Z = V_{Z1} + V_{Z2}$

$= 6 + 6$

$\therefore V_Z = 12\,V$

The equivalent circuit of zener regulator is



i) Maximum load current, $I_{L(max)} = \dfrac{V_L}{R_{L(min)}} = \dfrac{V_Z}{R_{L(min)}}$

$= \dfrac{12}{200} = 60\,mA$

And, Minimum load current, $I_{L(min)} = \dfrac{V_L}{R_{L(max)}} = \dfrac{V_Z}{R_{L(max)}}$

$= \dfrac{12}{1000} = 12\,mA$

ii> $P_{z(min)} = V_z \cdot I_{z(min)}$

And $P_{z(max)} = V_z \cdot I_{z(max)}$

Now,

$$I = \frac{V_{in} - V_z}{R_s} = \frac{15-12}{100} = 30\,mA$$

$$I_{z(min)} = I - I_{L(max)} = 30 - 60 = -30\,mA$$

$$I_{z(max)} = I - I_{L(min)} = 30 - 12 = 18\,mA$$

$$\therefore P_{z(min)} = 12 \times (-30) = -360\,mW$$

$$\therefore P_{z(max)} = 12 \times (18) = 216\,mW$$

iii> Voltage drop across series resistance.

$$V_R = I \cdot R_s$$
$$= 30 \times 100$$
$$= 3000 \times 10^{-3}$$
$$\therefore V_R = 3\,V \quad (\because V_{in} - V_z = 15 - 12 = 3)$$

iv> $$R_{L(min)} = \frac{V_L}{I - I_{z(min)}} = \frac{V_z}{30-(-30)} = \frac{12}{60} = 200\,\Omega$$

$$R_{L(max)} = \frac{V_z}{I - I_{z(max)}} = \frac{12}{30-18} = \frac{12}{12} = 1\,k\Omega$$

1\. a. If the reverse saturation current of diode is 5nA at 35°c. Find the reverse saturation current at 235°c.

Sol Let T_1 and T_2 be the temperature of 35°c and 235°c respectively.

i.e. $T_1 = 35°c$ & $T_2 = 235°c$

Let I_1 and I_2 be the reverse saturation current at 35°c and 235°c respectively.

i.e. $I_1 = 5nA$

$I_2 = ?$

we know that

$$I_2 = \left(2^{\frac{T_2 - T_1}{10}}\right) . I_1$$

$$= 2^{\frac{235-35}{10}} \times 5 \times 10^{-9}$$

$$\therefore I_2 = 5.24\, mA$$

Thus the reverse saturation current at 235°c is 5.24 mA Ans.

2\. For silicon diode, the reverse saturation current is 20μA at 27°c. What will be the value of reverse saturation current at 87°c?

Sol Let I_1 and I_2 be the reverse saturation current at temperature $T_1 = 27°c$ and $T_2 = 87°c$ respectively.

we know that, $I_1 = 20\mu A$; $I_2 = ?$

$$I_2 = \left(2^{\frac{T_2 - T_1}{10}}\right) . I_1$$

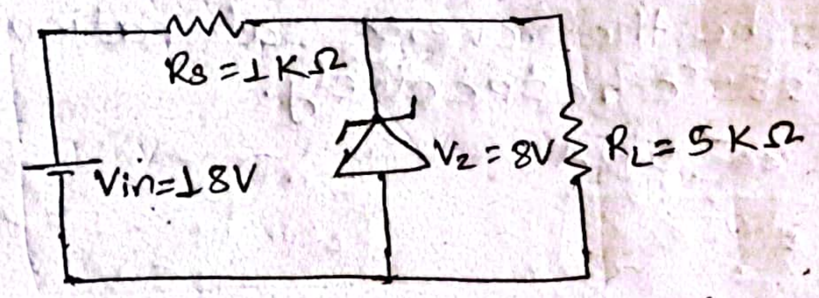
$$= \left(2^{\frac{87-27}{10}}\right) . 20 \times 10^{-6}$$

$$\therefore I_2 = 1.28 mA$$

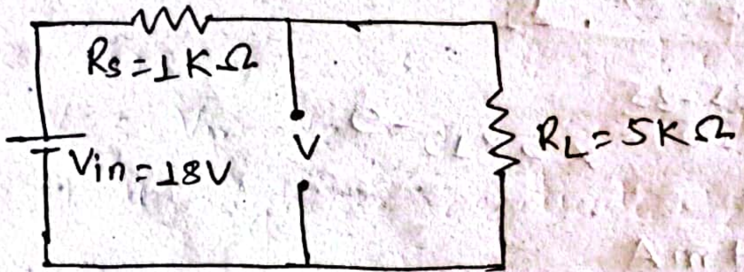
Thus the required value of reverse saturation current at 27°c is 1.28mA. Ans

1. For the circuit shown below, find
i> load voltage, V_L.
ii> zener current, I_z
iii> Power across zener diode, P_z.

Sol The given circuit is



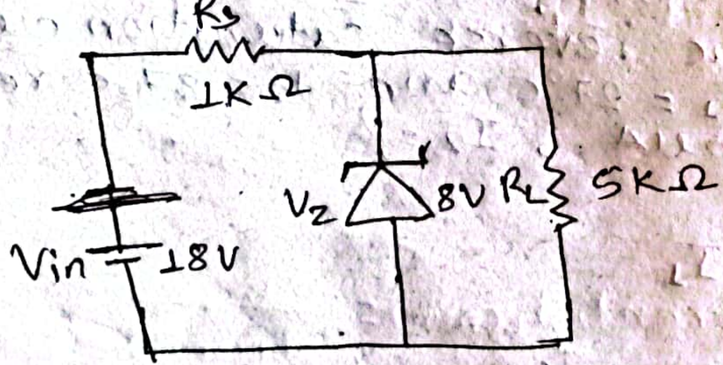
calculating open circuit voltage,



$$\therefore V = V_{in} \times \frac{R_L}{R_S+R_L} = 18 \times \frac{5}{1+5} = 30V$$

Since $V > V_z$. So the zener diode is ON.

The equivalent circuit is



i> Load voltage, $V_L = V_z = 8$ V Ans

Here,

ii> Zener current, I_Z:

We know,

$$I = I_L + I_Z$$

$$\therefore I_Z = I - I_L$$

Now,

$$I_Z = I - I_L = \frac{V_{in} - V_Z}{R_S} - \frac{V_Z}{R_L}$$

$$= \frac{18-8}{1000} - \frac{8}{5000} = \frac{10 \times 5 - 8}{5000}$$

$$\therefore I_Z = 8.4\,mA \quad \underline{\underline{Ans}}$$

iii> Power across zener diode, P_Z:

$$P_Z = V_Z \cdot I_Z$$

$$= 8 \times 8.4$$

$$\therefore P_Z = 67.2\,mW \quad \underline{\underline{Ans}}$$

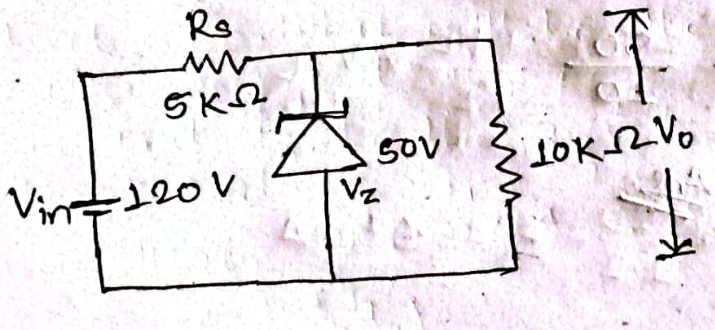
2. For the circuit shown in figure below find

i> Output voltage, V_o

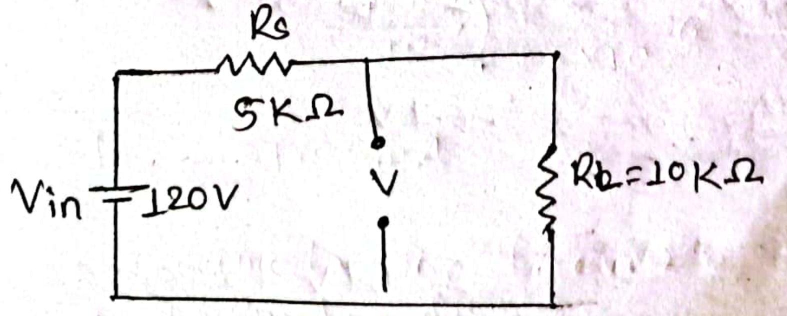
ii> Voltage across series resistance, R_s

iii> Current through zener diode, I_Z

Sol The given circuit is:-



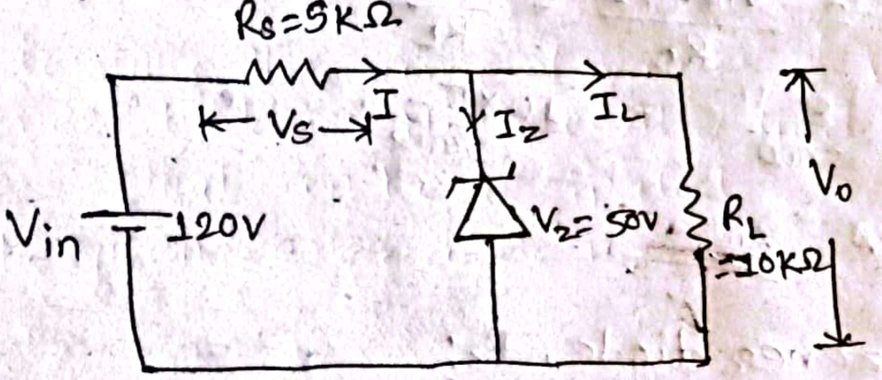
Calculating open circuit voltage,



$\therefore V = V_{in} \times \frac{R_L}{R_S + R_L} = 120 \times \frac{10}{10+5} = 80V$

Since $V > V_Z$. So the zener diode is ON.

The equivalent circuit is



i> Output voltage, $V_o = V_Z = 50V$ Ans

ii> voltage across series resistance, V_S:

$V_S = V_{in} - V_Z = 120 - 50 = 70V$ Ans

iii> Current through zener diode, I_Z:

we have,

$I = I_L + I_Z$

$\Rightarrow I_Z = I - I_L$

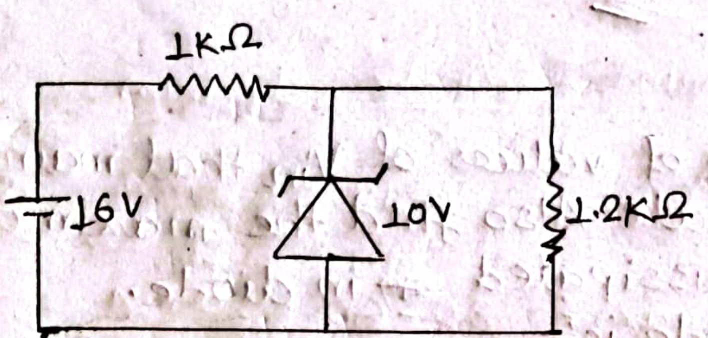
$\Rightarrow I_Z = \cancel{\frac{V_{in} - V_Z}{R_S}} \frac{V_S}{R_S} - \frac{V_Z}{R_L}$

$\Rightarrow I_Z = \frac{70}{5} - \frac{50}{10}$

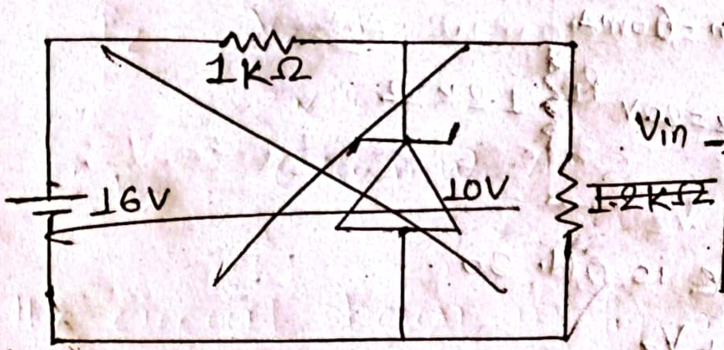
$\therefore I_Z = 9mA$ Ans

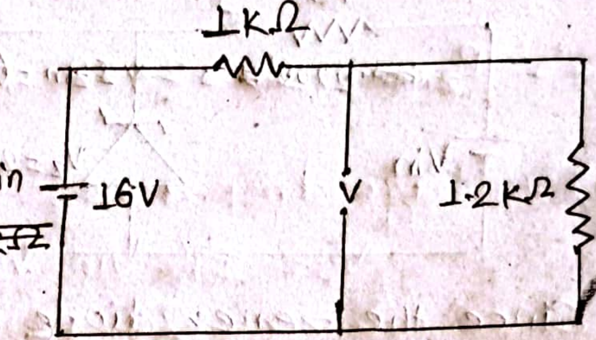
3. For the zener diode network shown in figure, determine V_L, V_R, I_Z and P_Z.

Sol. The given network is



calculating open circuit voltage



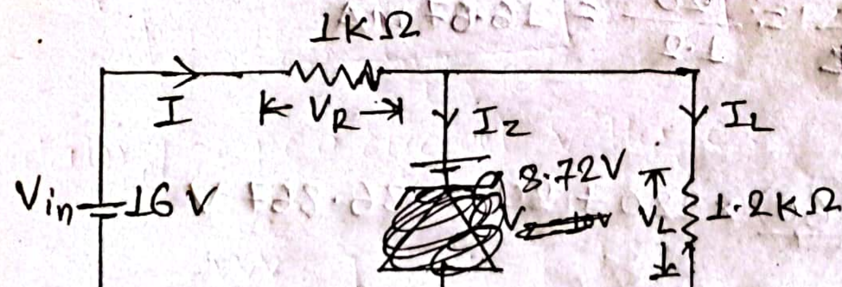


$$\therefore V = \cancel{80}\, V_{in} \times \frac{R_L}{R_S + R_L} = \frac{16 \times 1.2}{1 + 1.2} = 16 \times \frac{1.2}{1+1.2}$$

$$= 8.72\,V$$

Since $V < V_Z$, so the zener diode is OFF.

Now, The equivalent circuit is :-



i) $V_L = V_Z = 10\,V$ Ans

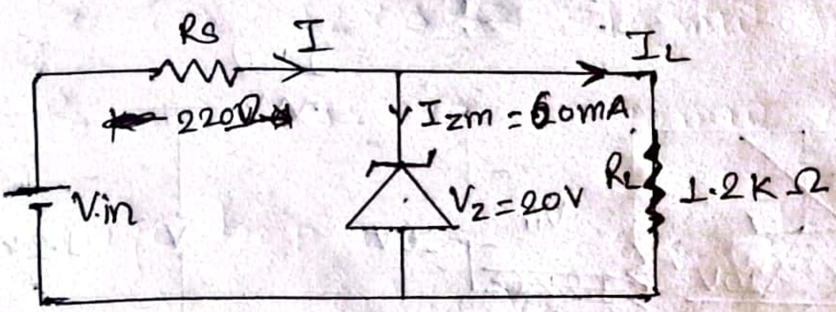
ii> $V_R = V_{in} - V_z = 16 - 8.72 = 7.28\ V$ Ans

iii> $I_Z = 0\ A$ Ans

iv> $P_Z = V_Z \cdot I_Z = 0\ W$ Ans

4. Determine the range of values of V_{in} that maintain zener diode is ON state. Also find the maximum power that can be dissipated in diode.

Sol: The given zener diode is:



Since the zener diode is ON. so

$$V_{in(max)} = I R_S + V_Z$$

$$\therefore V_{in(max)} = (I_L + I_{zm}) \times R_S + 20 \quad \text{---} (1)$$

Also,

$$V_Z = V_{in(min)} \times \frac{R_L}{R_S + R_L}$$

$$\therefore V_{in(min)} = \frac{R_S + R_L}{R_L} \times V_Z \quad \text{---} (2)$$

we have,

$$I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L} = \frac{20}{1.2} = 16.67\ mA$$

from (1),

$$V_{in(max)} = \frac{(16.67 + 60)}{1000} \times 220 + 20 = 36.867\ V$$

$$V_{in(min)} = \left(\frac{220 + 1200}{1200}\right) \times 20 = 23.67\ V$$

Hence, the range of values of Vin is 23.67V to 36.867V. Ans

Maximum power dissipated in zener diode is:

$$P_{z(max)} = V_z \cdot I_{zm}$$
$$= 20 \times 60$$
$$= 1.2\ W$$ Ans

5. A zener regulator has 12V zener voltage with variable load resistance. Calculate

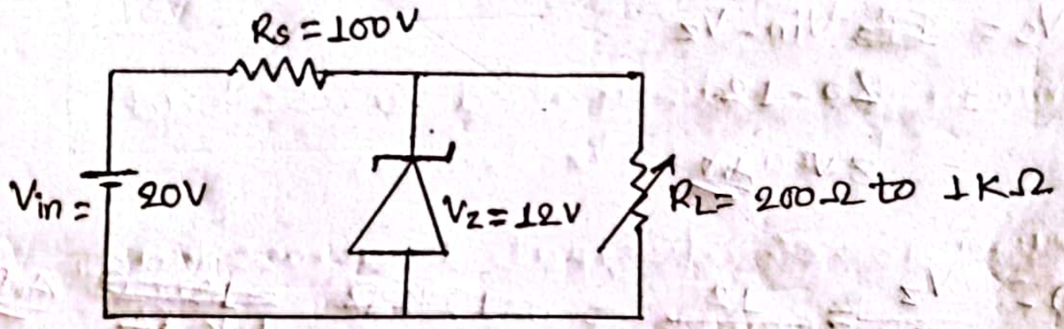
i) Maximum and minimum load current

ii) Maximum and minimum power dissipated across load.

iii) Voltage drop across series resistance

iv) Maximum value of load resistance to ensure that zener diode is ON.

The given zener diode circuit is:



~~Calculating~~

Since the zener diode is ON.

$V_{in} = 20\ V$

$R_s = 100\ V$

$V_z = 12\ V$

$R_{L(min)} = 200\ \Omega$

$R_{L(max)} = 1\ K\Omega = 1000\ \Omega$

i) Maximum load current, $I_{L(max)}$:

we know, $I_{L(max)} = \frac{V_L}{R_{L(min)}} = \frac{V_z}{R_{L(min)}} = \frac{12}{200} = 60\ mA$

And,

Minimum load current, $I_{L(min)} = \frac{V_z}{R_{L(max)}} = \frac{12}{1000} = 12\ mA$

ii> Maximum power dissipated across load,

$P_{z(max)} = V_z \cdot I_{z(max)}$

$P_{z(min)} = V_z \cdot I_{z(min)}$

Now,

$I_{z(max)} = I - I_{L(min)} = \frac{V_{in}-V_z}{R_s} - I_{L(min)}$

$\therefore I_{z(max)} = \frac{20-12}{100} - 12 = 80-12 = 68\,mA$ Ans

Also, $I_{z(min)} = I - I_{L(max)} = 80-60 = 20\,mA$

Now,

$\therefore P_{z(max)} = 12 \times 68 = 816\,mW$ Ans

$\therefore P_{z(min)} = 12 \times 20 = 240\,mW$ Ans

iii> Voltage drop across series resistance, V_s:

$V_s = V_{in} - V_z$

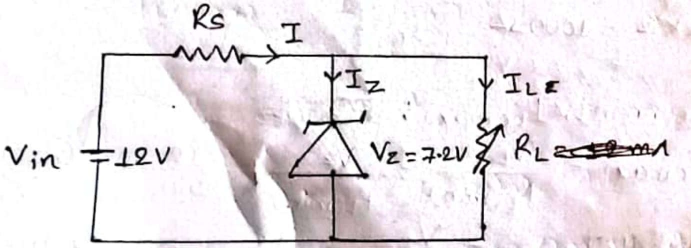
$= 20 - 12$

$= 8V$ Ans

iv> $R_{L(max)} = \frac{V_z}{I - I_{zm}} = \frac{12}{80mA - 68} = 1\,k\Omega$ Ans

6. For the given circuit, load current varies from 12mA to 100 mA. Find the value of series resistance R_s to maintain 7.2V across the load. Maximum zener diode current is 10mA.

Sol The given circuit is:



Given:

$V_{in} = 12V$

$V_z = 7.2V$

$I_{L(min)} = 12mA$

$I_{L(max)} = 100mA$

$I_{z(min)} = 10\,mA$

$\therefore I = I_{L(max)} + I_{z(min)}$

$= 100 + 10$

$\therefore I = 110\,mA$

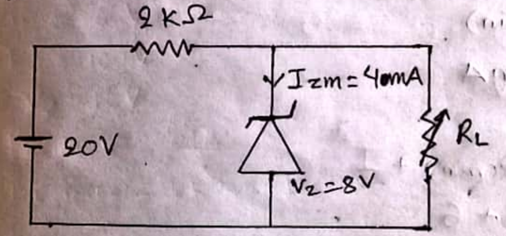
Now,

Series resistance, $R_s = \frac{V_{in}-V_z}{I} = \frac{12-7.2}{110}$

$= 43.63\,\Omega$ Ans

7. For the circuit given, find the maximum and minimum value of load resistor R_L required for safe turn 'ON' of zener diode. Also find the minimum and maximum power that can be dissipated across load.

Sol The given circuit is:



Given:-

$V_{in} = 20V$

$R_s = 2k\Omega$

$V_z = 8V$

$I_{zm} = 40mA$.

we know,

$$I = \frac{V_{in} - V_z}{R_s} = \frac{20-8}{2\times1000} = 6mA$$

i> Maximum load resistor, $R_{L(max)} = \frac{V_z}{I - I_{z(max)}}$

$$= \frac{8}{6-4}$$

$\therefore R_{L(max)} = 4k\Omega$ Ans

Also,

Minimum load resistor, $R_{L(min)} = \frac{V_z}{I}$

$$= \frac{8}{6}$$

$\therefore R_{L(min)} = 1.33k\Omega$ Ans

ii> Maximum power dissipated across load,

$P_{z(max)} = V_z \cdot I_{z(max)}$

Minimum power dissipated across load,

$P_{z(min)} = V_z \cdot I_{z(min)}$

Now, $I = I_{z(max)} + I_{L(min)}$

$\Rightarrow 6 = 4 + I_{L(min)}$

$\therefore I_{L(min)} = 2mA$

Also,

$I_{L(max)} = I_{z(min)} + I$

$= 0 + 6$

$\therefore I_{L(max)} = 6mA$

we have

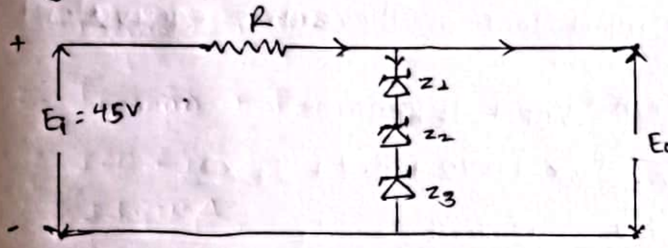
$\therefore P_{z(max)} = 8\times6 = 48mW$ Ans

$\therefore P_{z(min)} = 8\times0 = 0mW$ Ans

2021-Fall

1.b. What value of series resistance (R) is required when three 10 W, 10V, 1000mA zener diodes are connected in series to obtain a 30V regulated output from a 45V dc power source?

Sol: The given circuit is:-



$\therefore V_z = V_{z1} + V_{z2} + V_{z3}$

$= 10+10+10$

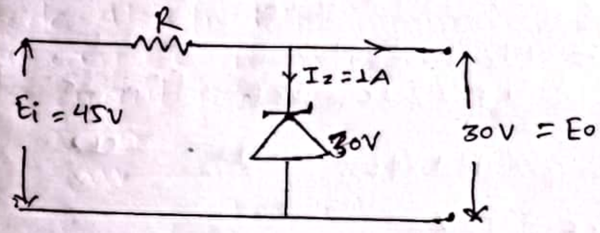
$\therefore V_z = 30V$

$\therefore P_z = 10+10+10$

$= 30W$

$\therefore I_z = 1000mA = 1A$

The equivalent circuit is:



$\therefore V_R = 45-30 = 15V$

$\therefore R = \frac{V_R}{I} = \frac{15}{I_z + I_L} = \frac{15}{1+0} = 15$

we know,

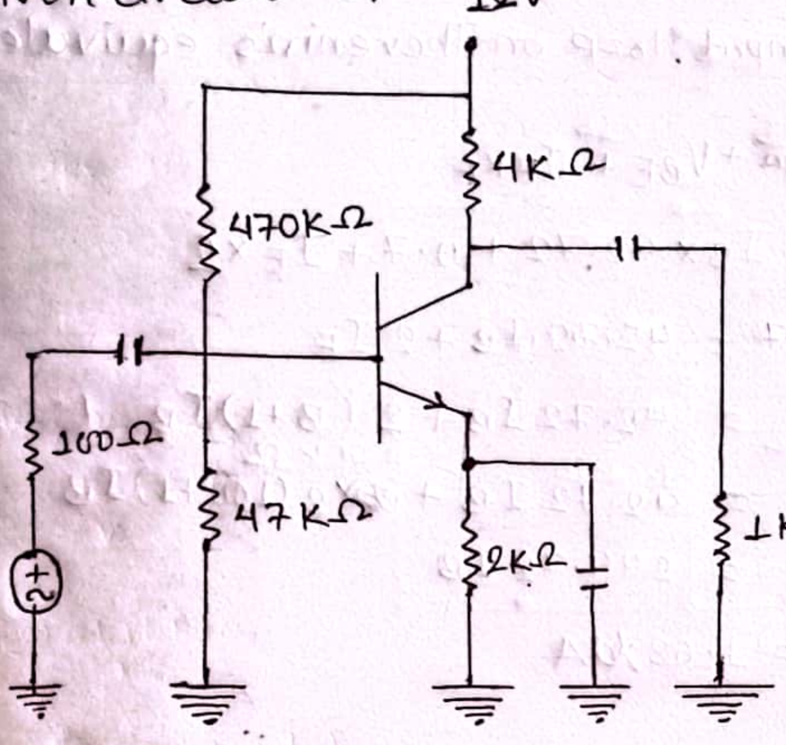
$V_z = V_o$

$\Rightarrow 30 = I_L R_L$

BJT (Prabin chaudhary)

Q.6. In the given silicon-biased circuit $\beta = 100$, draw the DC and AC load line and determine the operating point.

Sol The given circuit is:



The dc equivalent circuit and Thevenin's equivalent circuit are:

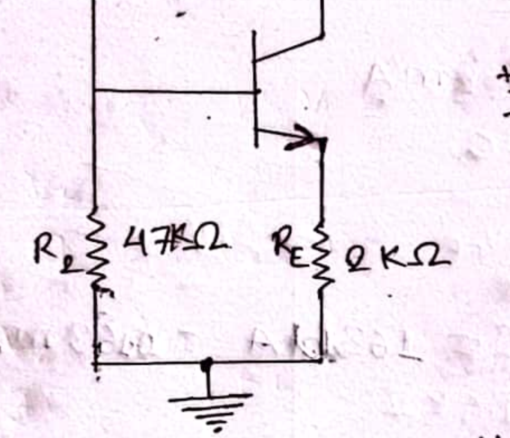


fig. dc equivalent circuit

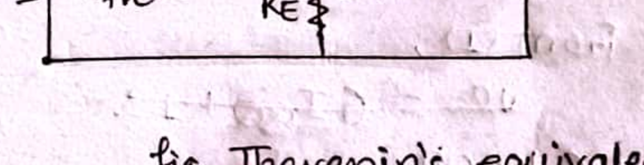


fig. Thevenin's equivalent circuit

Here,

$$R_{th} = R_1 // R_2 = \frac{470 \times 47}{470+47} = 42.72 K\Omega$$

$$V_{th} = V_{CC} \times \frac{R_2}{R_1+R_2} = 12 \times \frac{47}{470+47} = 1.09V$$

Applying KVL in input loop on Therenin's equivalent circuit, we get;

$$V_{th} = I_B \times R_{th} + V_{BE} + I_E R_E$$

$$\Rightarrow 1.09 = I_B \times 42.72 + 0.7 + I_E \times 2$$

$$\Rightarrow 1.09 - 0.7 = 42.72 I_B + 2 I_E$$

$$\Rightarrow 0.4 = 42.72 I_B + 2(\beta+1) I_B \quad (\because I_E = (\beta+1) I_B)$$

$$\Rightarrow 0.4 = 42.72 I_B + 2(100+1) I_B$$

$$\Rightarrow 0.4 = 244.72 I_B$$

$$\therefore I_B = 1.63 \mu A$$

Applying KVL in output loop ~~we get~~ on Therenin's equivalent circuit, we get;

$$V_{CC} = 4 I_C + V_{CE} + I_E R_E$$

$$\Rightarrow 12 = 4 I_C + 2 I_E + V_{CE}$$

$$\Rightarrow 12 = 4 I_C + 2 I_C + V_{CE} \quad (\because I_C \approx I_E)$$

$$\therefore 12 = 6 I_C + V_{CE} \quad \text{--- (1)}$$

when $I_C = 0$, $V_{CE} = 12V$

when $V_{CE} = 0$; $I_C = 2mA$

~~from (1),~~

~~$12 = 6 I_{CQ} + 12$~~

we know,

$$I_{CQ} = \beta I_B = 100 \times 1.63 = 163 \mu A = 0.163 mA$$

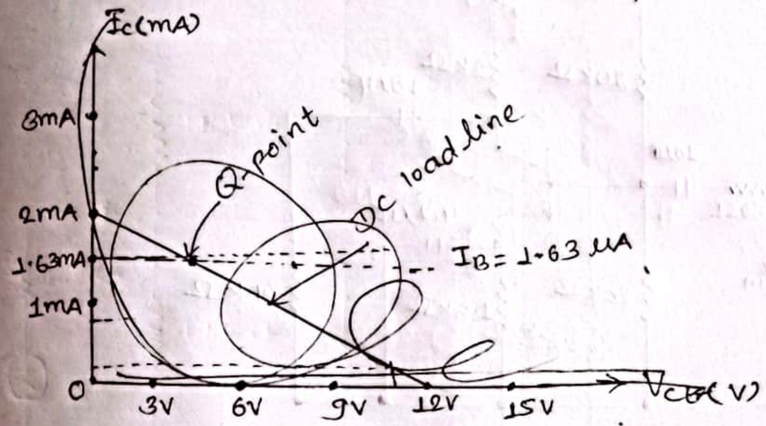
$\therefore$ from (1),

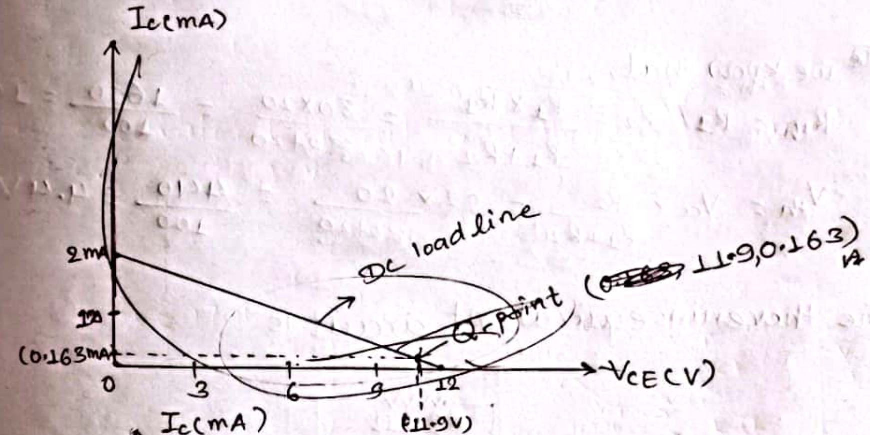
$$12 = 6 \times 0.163 + V_{CE}$$

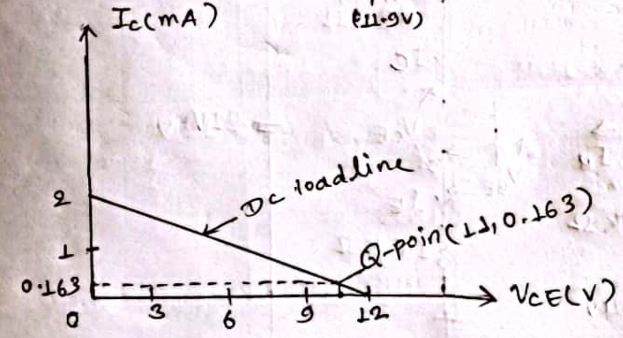
$$\Rightarrow 12 - 0.978 = V_{CE}$$

$\therefore V_{CEQ} = 11$

The dc load line is the straight line joining two points (12V, 0) and (2mA, 0) as shown in output characteristics of CE transistor configuration.

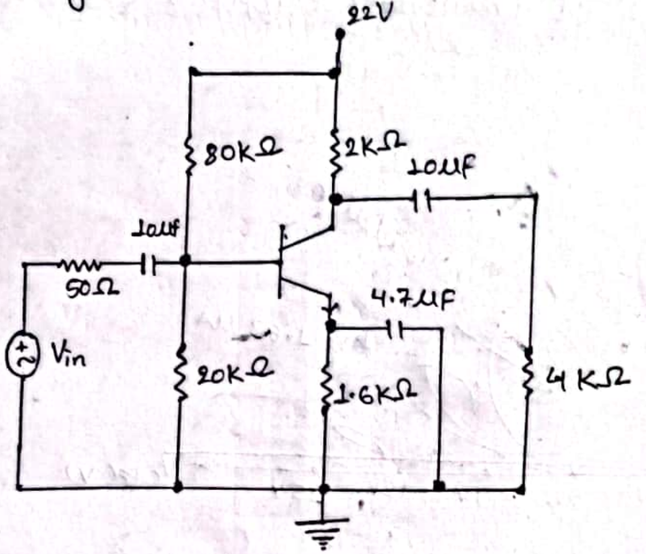






26. For the BJT common emitter circuit shown below, determine the operating point and locate it in the load line. Assume $\beta = 100$.

Sol: The given CE circuit is:

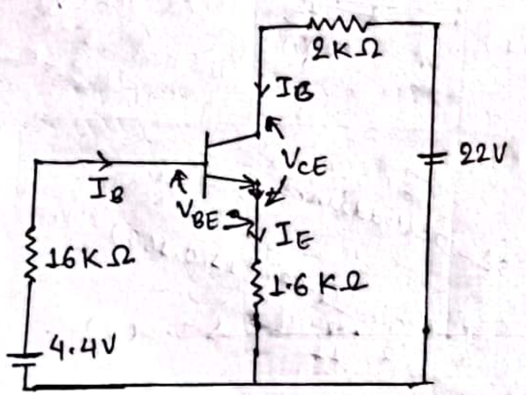


We know that,

$$R_{th} = R_1 // R_2 = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{80 \times 20}{80+20} = \frac{1600}{100} = 16K\Omega$$

$$V_{th} = V_{cc} \times \frac{R_2}{R_1+R_2} = \frac{22 \times 20}{80+20} = \frac{440}{100} = 4.4V$$

The thevenin's equivalent circuit is



Applying KVL in input loop, we get:-

$$4.4 = 16 \times I_B + V_{BE} + 1.6 \times I_E$$

$$\Rightarrow 4.4 = 16 I_B + 0.7 + 1.6 \times (100+1) I_B$$

$$\Rightarrow 4.4 - 0.7 = 177.6 I_B$$

$$\Rightarrow \frac{3.7}{177.6 \times 1000} = I_B$$

$$\therefore I_B = 20.8 \mu A$$

Applying KVL in output loop, we get:

$$22 = 2 \times I_C + V_{CE} + 1.6 \times I_E$$

$$\Rightarrow 22 = 2I_C + 1.6 I_C + V_{CE} \quad (\because I_C \approx I_E)$$

$$\Rightarrow 22 = 3.6 I_C + V_{CE} \quad \text{——— (1)}$$

when $I_C = 0$, $V_{CE} = 22$

when $V_{CE} = 0$, $I_C = 13.75 mA$

we know,

$$I_{CQ} = \beta I_B = 100 \times 20.8 \times 10^{-6} = 2.08 mA$$

from (1),

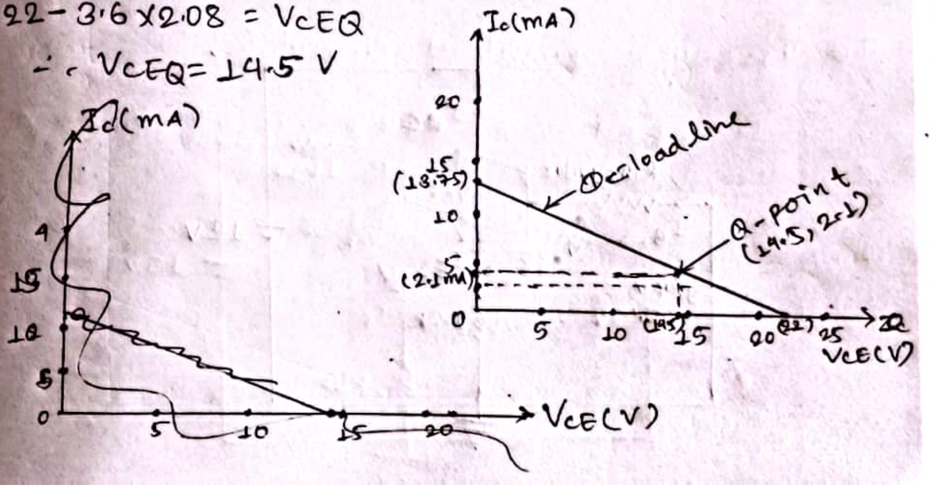
~~$V_{CEQ} = 22 - 3.6 \times 2.08 \times 10^{-3}$~~

$$\therefore I_E = I_C + I_B = 2.08 \times 10^{-3} + 20.8 \times 10^{-6} = 2.1 mA$$

from (1),

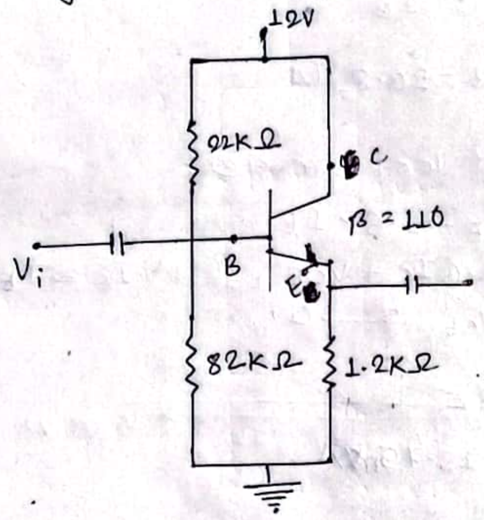
$$22 - 3.6 \times 2.08 = V_{CEQ}$$

$$\therefore V_{CEQ} = 14.5 V$$



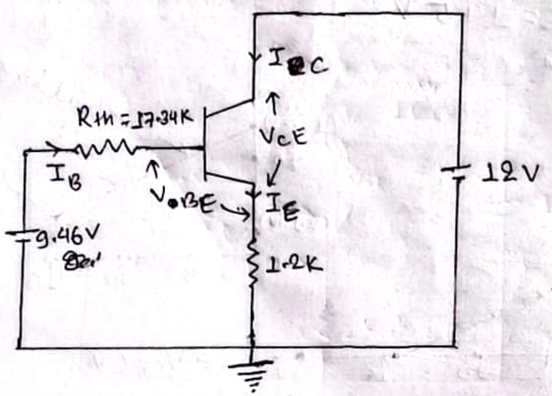
2.b. For the emitter follower network of figure below
i) Find I_B, I_C and I_E
ii) Determine V_B, V_C, and V_E.
iii) calculate V_{BC}

Sol: The given emitter follower network is shown below



Now,
$$R_{th} = \frac{22 \times 82}{22+82} = 17.34\,K\Omega$$
$$V_{th} = 12 \times \frac{82}{22+82} = 9.46\,V$$

The Thevenin's equivalent circuit is:



Now Applying KVL in input loop, we get:
$$9.46 = 17.34\,I_B + V_{BE} + 1.2\,I_E$$
$$\Rightarrow 9.46 = 17.34\,I_B + 0.7 + 1.2(\beta+1)I_B$$
$$\Rightarrow 9.46 - 0.7 = 17.34\,I_B + 1.2(110+1)\,I_B$$
$$\Rightarrow 8.76 = 150.54\,I_B$$
$$\therefore I_B = 0.58\,\mu A \quad \text{Ans.}$$

Also we know,
$$I_C = \beta I_B = 110 \times 0.58 \times 10^{-6} = 0.64\,\mu A \quad \text{Ans}$$
$$I_E = (\beta+1)\,I_B = 111 \times 0.58 \times 10^{-6} = 0.64\,\mu A \quad \text{Ans}$$

Now,
$$V_B = V_{th} = 9.46\,V \quad \text{Ans}$$
$$V_C = I_C R_C = I_C \times 0 = 0\,V \quad \text{Ans}$$
$$V_E = I_E R_E = 0.64 \times 10^{-6} \times 1.2 \times 10^{3} = 0.768\,mV \quad \text{Ans}$$

2016.Spring

2.c. An NPN silicon transistor having a normal β of 100 is to be used in a CE configuration with $V_{cc} = 15V$. The Q... is to be $I_c = 4mA$ and $V_{CE} = 12V$. Now design the circuit diagram.

Sol Given;

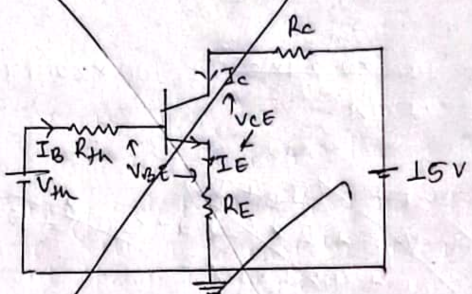
$\beta = 100$

$V_{cc} = 15V$

$I_{cQ} = 4mA$

$V_{CEQ} = 12V$

The ~~required circuit~~ diagram will be
Thevenin's eqvt



Apply KVL in output loop, we get;

$$15 = I_cR_c + V_{CE} + I_ER_E$$

we have,

$$I_c = \beta I_B$$

$$\Rightarrow I_B = \frac{I_c}{\beta} = \frac{4\times10^{-3}}{100} = 0.4\,\mu A$$

$$\therefore I_E = (\beta+1) I_B = 101 \times 0.4\times10^{-6} = 4.04\,mA$$

Assuming, $V_T = 26\,mV$

$$g_m = \frac{I_c}{V_T} = \frac{4}{26} = 0.153.$$

~~we know~~, since $V_{cc} \geq 15$. so we use guideline 2 (1/10th rule) for the design of bias circuit.

From (1/10th) rule,

$$V_E = I_ER_E = \frac{V_{cc}}{10}$$

$$\Rightarrow 4.04 R_E = \frac{15}{10} = 1.5$$

$$\therefore R_E = \frac{1.5}{4.04\times10^{-3}} = 371.28\,\Omega = 0.371\,k\Omega$$

Also,

$$V_{Rc} = \frac{4}{10} V_{cc} = \frac{4}{10} \times 15 = 6.$$

$$\Rightarrow I_cR_c = 6$$

$$\Rightarrow R_c = \frac{6}{4\times10^{-3}} = 1500 = 1.5\,k\Omega$$

Using stiff voltage divider biasing.

$$R_2 = 0.01\,\beta R_E = 0.01\times100\times0.371\times10^3$$

$$\therefore R_2 = 0.371\,k\Omega$$

For stiff biasing,

$$I_{R12} = I_E = 4.04\,mA$$

where, I_{R12} is the series current flowing through R_1 and R_2.

The application of KVL resultss.

$$V_{cc} = I_{R12}(R_1 + R_2)$$

$$\Rightarrow 15 = 4.04\times10^{-3}(0.371 + R_1)$$

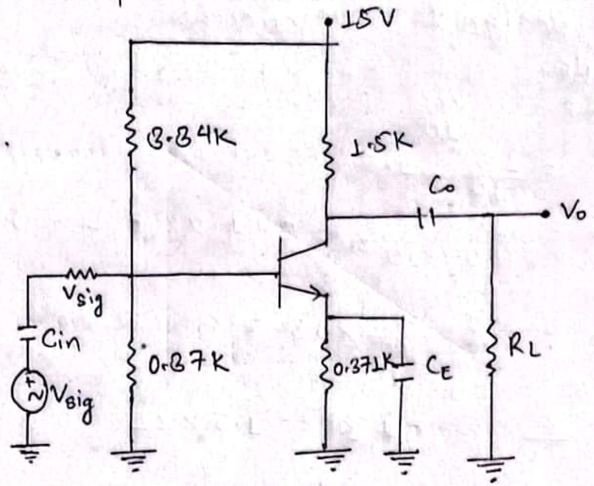
$$\Rightarrow \frac{15}{4.04\times10^{-3}} = 0.371 + R_1$$

$$\Rightarrow R_1 = 3712.871 - 0.371\times1000$$

$$= \cancel{3341.87\,k\Omega}$$

$$\therefore R_1 = 3.34\,k\Omega$$

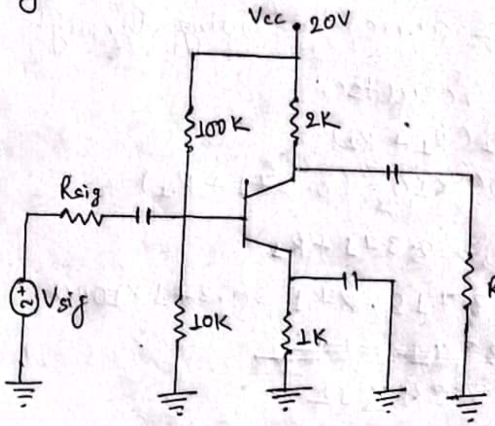
Thus the complete common emitter circuit is:



2b. (2015-Spring)

For the circuits shown below, draw the dc load line and find the operating point. Also find the stability S of the circuit. (Assume silicon transistor $\beta = 100$).

Sol: The given circuit shown below is:

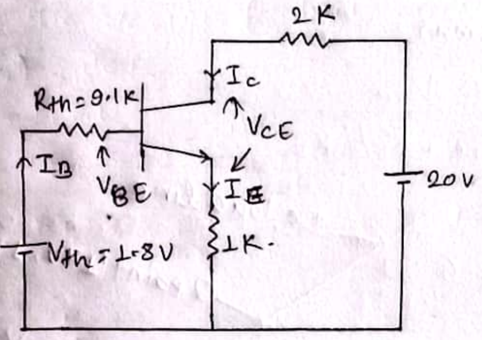


we know that

$$R_{th} = \frac{10 \times 100}{10 + 100} = 9.1\,K\Omega$$

$$V_{th} = V_{cc} \times \frac{10}{10+100} = \frac{20 \times 10}{110} = 1.8\,V$$

The Thevenin's equivalent circuit is:-



Applying KVL in input loop, we get:

$$1.8 = 9.1 I_B + V_{BE} + I_E$$

$$\Rightarrow 1.8 = 9.1 I_B + 0.7 + 101 I_B$$

$$\Rightarrow 1.8 - 0.7 = 110.1 I_B$$

$$\therefore I_B = 9.99 \approx 10\,\mu A$$

Applying KCL in output loop, we get:

$$20 = 2 I_C + V_{CE} + I_E$$

$$\Rightarrow 20 = 2 I_C + V_{CE} + I_C \quad (\because I_C \approx I_E)$$

$$\Rightarrow 20 = 3 I_C + V_{CE} \text{ ——— (i)}$$

when $I_C = 0$, $V_{CE} = 20V$

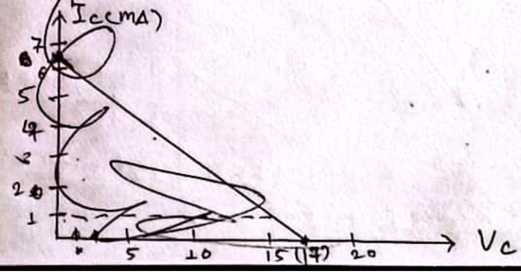
when $V_{CE} = 0$, $I_C = 6.67\,mA$

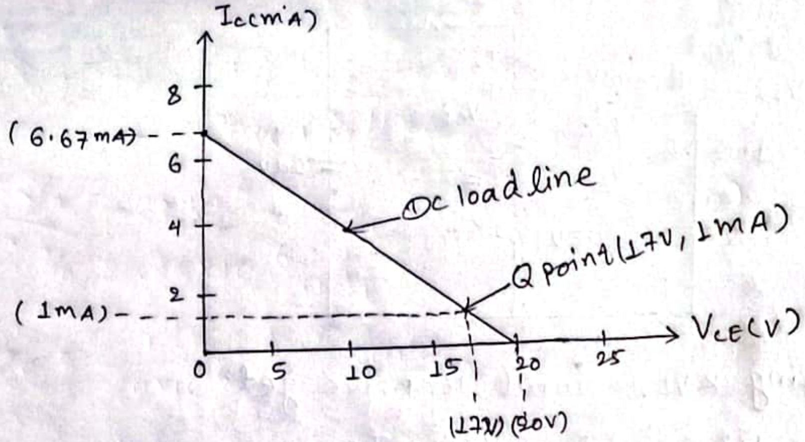
we know,

$$I_{CQ} = \beta I_B = 100 \times 10 \times 10^{-6} = 1mA$$

$$V_{CEQ} = 20 - 3 \times 1 = 17\,V$$

Now, The dc load line and operating point is shown in output characteristics of CE configuration.

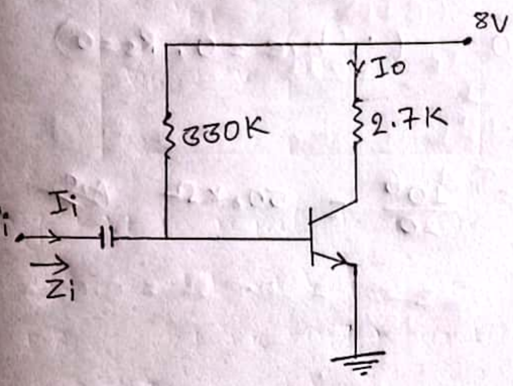




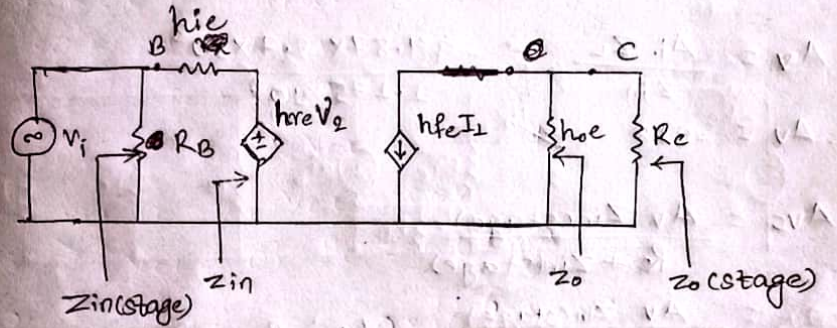
Small signal frequency model (Prabin)

4.b. Find A_{vs}, A_v, Z_o, Z_{in} of the following circuit diagram using h-parameter. (Given: $h_{fe} = 100$, $h_{ie} = 1.175K$, $h_{oe} = 20\mu A$)

Sol The given circuit diagram is:



The h-model of given circuit diagram is:



Here,

$Z_L = R_c = 2.7K\Omega$

$R_B = 330K\Omega$

i) $Z_{in} = h_{ie} - \dfrac{h_{fe}\, h_{re}}{h_{oe} + Y_L} = 1.175K\Omega - 0$ (∵ $h_{re} = 0$)

$= 1.175K\Omega$ Ans

ii) $Z_o = \frac{1}{Y_o}$

we know,

$$Y_o = h_{oe} - \frac{h_{fe}\, h_{re}}{h_{ie} + R_s // R_B}$$

$\Rightarrow Y_o = 20\times10^{-6} - 0$ $\quad(\because h_{re} = 0,\ R_s = 0)$

$\therefore Y_o = 20\times10^{-6}$

Now,

$$Z_o = \frac{1}{20\times10^{-6}} = \frac{10^6}{20} = 50\,K\Omega \quad \text{Ans}$$

iii) $A_i = \frac{-h_{fe}}{1+h_{oe}Z_L}$

$$= \frac{-100}{1+20\times10^{-6}\times2.7\times10^3}$$

$\therefore A_i = -94.87$ Ans

iv) $A_v = \frac{A_i \cdot Z_L}{Z_{in}} = \frac{-94.87\times2.7\times10^3}{1.175\times10^3}$

$\therefore A_v = -218$ Ans

v) $A_{vs} = \frac{A_v\, Z_{in(stage)}}{R_s + Z_{in(stage)}}$

$= \frac{A_v\, Z_{in(stage)}}{Z_{in(stage)}}$ $\quad(\because R_s = 0)$

$\therefore A_{vs} = A_v = -218$ Ans

2017-Spring

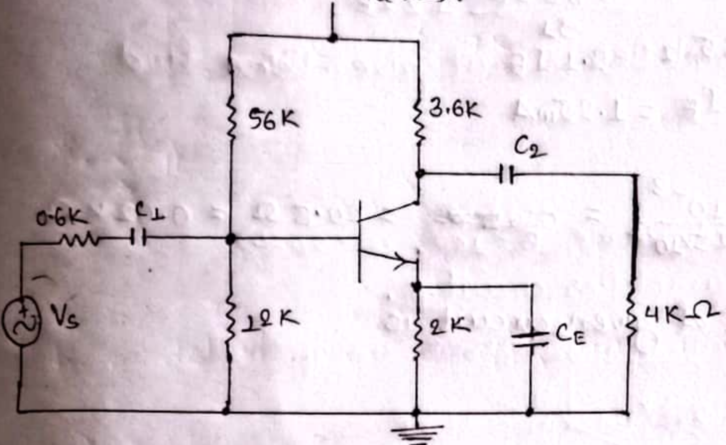
Q.b. For the given CE amplifier, find the following using re model

i. input impedance

ii) output impedance

iii) voltage gain (Assume $B = 90$, $V_{cc} = +19V$)

Sol The given CE amplifier is:-



~~we know,~~

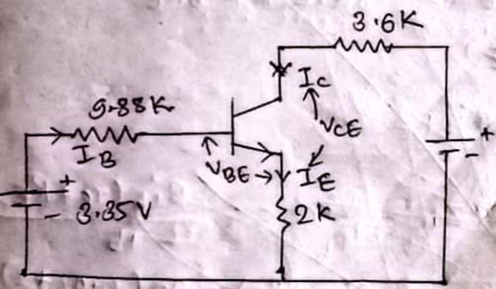
~~$r_e = \frac{26mV}{I_E}$~~

we have

$R_{th} = \frac{56\times12}{56+12} = 9.88\,K\Omega$

$V_{th} = V_{cc}\times\frac{12}{56+12} = \frac{19\times12}{56+12} = 3.35\,V$

The Thevenin's equivalent circuit is:-



Applying KVL in input loop, we get,

$$8.35 = 9.88 I_B + V_{BE} + 2 I_E$$

$$\Rightarrow 8.35 = 9.88 I_B + 0.7 + 2 I_E$$

$$\Rightarrow 8.35 - 0.7 = \frac{9.88 I_E}{(\beta+1)} + 2 I_E$$

$$\Rightarrow 2.65 = \frac{9.88 I_E}{91} + 2 I_E$$

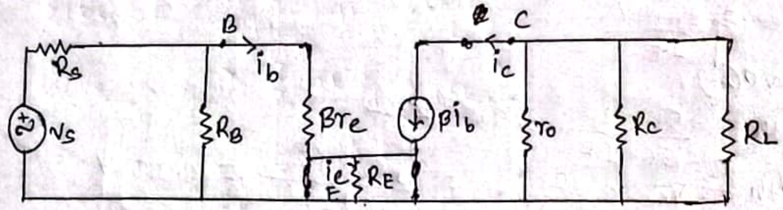
$$\Rightarrow 2.65 = 2.1 I_E$$

$$\therefore I_E = 1.25 mA$$

Now,

$$r_e = \frac{26 \times 10^{-3}}{1.25 \times 10^{-3}} = 20.8 \Omega = 0.02 K\Omega$$

The re-model of given circuit is



Now,

$$V_{in} = i_b \beta r_e \times i_e R_E = i_b \beta r_e + \beta i_b R_E$$

$$\therefore V_{in} = \beta i_b (r_e + R_E)$$

$$V_o = i_e R_E = \beta i_b R_E$$

i) Input impedance, $Z_{in} = \frac{V_{in}}{i_b} = \frac{\beta i_b (r_e + R_E)}{i_b}$

$$= \beta(r_e + R_E) = 90(0.02 + 2)$$

$$\therefore Z_{in} = 181.8 K\Omega$$

ii) Output impedance, $Z_o = R_c // R_L$

$$= \frac{R_c \times R_L}{R_c + R_L}$$

$$= \frac{3.6 \times 4}{3.6 + 4}$$

$$\therefore Z_o = 1.89 K\Omega$$

iii) Voltage gain, $A_v = \frac{V_o}{V_{in}} = \frac{-\beta i_b R_E}{\beta i_b (r_e + R_E)}$

$$= \frac{-R_E}{r_e + R_E}$$

$$= \frac{-2}{0.02 + 2}$$

$$\therefore A_v = 0.99 \approx 1$$

i) Input impedance, $Z_{in} = R_B // r_{in}$

we know,

$$r_{in} = \frac{\beta i_b (r_e + R_E)}{i_b} = \beta(r_e + R_E) = 90(0.02 + 2)$$

$$= 181.8 K\Omega$$

$$\therefore r_{in} = 181.8 K\Omega$$

Now,

$$Z_{in} = R_B // r_{in} = R_{th} // r_{in} = \frac{R_{th} \times r_{in}}{R_{th} + r_{in}}$$

$$\therefore Z_{in} = \frac{9.88 \times 181.8}{9.88 + 181.8} = 9.37 K\Omega$$

ii) Output impedance, $Z_o = r_o // R_c // R_L$

$$= R_c // R_L \quad (\because r_o = 0)$$

$$= \frac{R_c \times R_L}{R_c + R_L}$$

$$= \frac{3.6 \times 4}{3.6 + 4} = 1.89$$

$$\therefore Z_o = 1.89 K\Omega$$

iii> Voltage gain, $A_v = \frac{-V_o}{V_{in}} = \frac{-R_E}{r_e + R_E}$

$= \frac{-2}{0.02+2}$

$\therefore A_v = -0.199$ Ans

Q.a. For the amplifier circuit given below, find

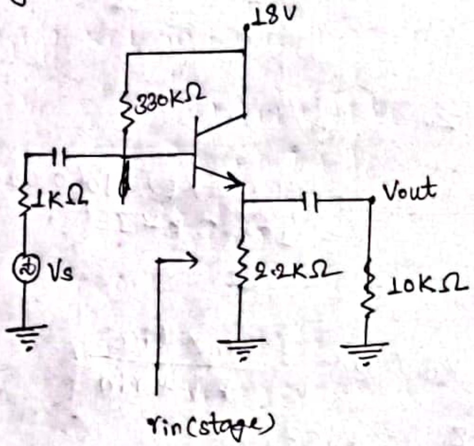
i> r_{in}

ii> r_{out}

iii> A_v

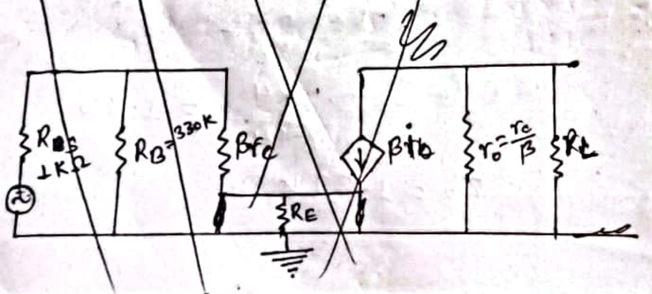
iv> A_{vs} (Assume $\beta = 100$, $r_e = 20\Omega$)

Sol The given amplifier circuit is;

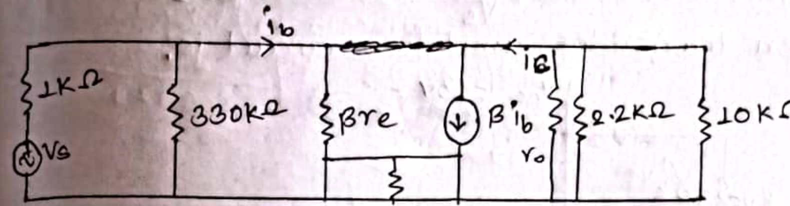


~~Applying KVL in~~

The re model of given circuit is;



The re-model of given circuit is:



i> $r_{in} = 1 // 330 // \beta r_e = \frac{1 \times 330}{1+330} // (100)(20)\times 10^{-3}$

$= 0.62\,K\Omega$

ii> $r_o = 2.2 // 10 = \frac{2.2 \times 10}{2.2 + 10} = 1.8\,K\Omega$

iii> $A_v = -\frac{R_E}{r_e + R_E} = -\frac{2.2 \times 10^3}{20 + 2.2 \times 10^3} = 0.99 = 1$

iv> $A_{vs} =$

i> $r_{in} = \beta(r_e + R_E) = 100(20\times10^{-3} + 2.2) = 222\,K\Omega$

ii> $r_{in}(stage) = r_{in} // R_B = \frac{222 \times 330}{222 + 330} = 132.7\,K\Omega$

iii> $r_o(stage) = r_o // R_C = R_C = 0$ ($\because r_o \cong \infty$)

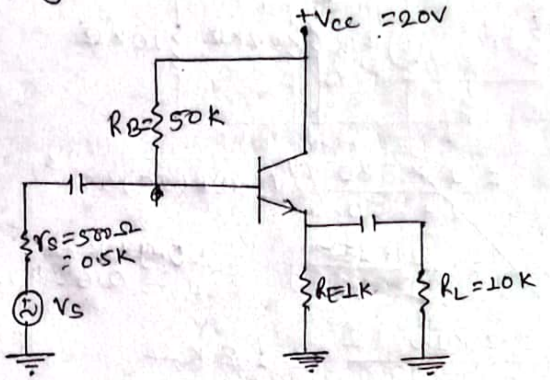
iv> $A_v = \frac{-R_C}{r_e + R_E} = -\frac{2.2}{20\times10^{-3} + 2.2} = 0$

v> $A_{vs} = A_v \left(\frac{r_{in}(stage)}{r_s + r_{in}(stage)}\right) \cdot \left(\frac{R_L + r_o stage}{R_L stage}\right)$

$= 0$ Ans

(*) For the transistor amplifier circuit shown below, draw its re-model and find r_{in}, $r_{in(stage)}$, $r_o(stage)$, A_v, $A_{vs} = \frac{V_L}{V_s}$. Assume $\beta = 100$.

Sol The given transistor amplifier circuit is:



Applying KCL in input loop, we get:

$$V_{cc} = I_B R_B + V_{BE} + I_E R_E$$

$$\Rightarrow 20 = 50 I_B + 0.7 + I_E$$

$$\Rightarrow 20 = 50 \times \frac{I_E}{(\beta+1)} + 0.7 + I_E$$

$$\Rightarrow 20 - 0.7 = \frac{50}{101} I_E + I_E$$

$$\Rightarrow 19.3 = 1.49 I_E$$

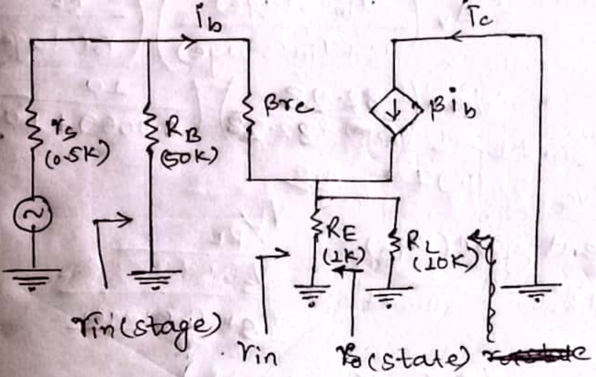
$$\therefore I_E = 12.95 mA$$

Now,

$$r_e = \frac{26mV}{I_E} = \frac{26mV}{12.95mA} = 2\Omega$$

$$\therefore r_L = R_E // R_L = 1 // 10 = \frac{1 \times 10}{1+10} = 0.91 K\Omega$$

The re-model of given circuit is:-



Now,

i> $r_{in} = \beta(r_e + r_L) = \beta(r_e + R_E // R_L) = \beta\left(r_e + \frac{R_E \times R_L}{R_E + R_L}\right)$

$$= 100\left(0.002 + \frac{1 \times 10}{1+10}\right) = 91.1 K\Omega \text{ Ans}$$

ii> $r_{in(stage)} = r_{in} // R_b = 91.1 // 50$

$$= \frac{91.1 \times 50}{91.1 + 50} = 32.28 K\Omega \text{ Ans}$$

iii> $r_o(stage) = R_E // \left(\frac{r_s // R_B}{\beta} + \beta r_e\right)$

$$= 1 // \left(\frac{0.5//50}{100} + 100 \times 2 \times 10^{-3}\right)$$

$$= 1 // \left(\frac{\frac{0.5 \times 50}{0.5+50} + 0.2}{100}\right)$$

$$= 1 // 0.00695$$

$$\therefore r_{o(stage)} \; 0.0069 K\Omega \text{ Ans}$$

iv> voltage gain, $A_v = \frac{r_L}{r_e + r_L} = \frac{0.91}{0.002 + 0.91} = 0.9978$ Ans

~~v> Current gain~~.

v) Overall voltage gain $(A_{vs}) = \frac{V_L}{V_s}$

$$= A_v \left(\frac{r_{in(stage)}}{r_s + r_{in(stage)}}\right)\left(\frac{R_L}{r_{o(stage)} + R_L}\right)$$

$$= 0.99\left(\frac{32.28}{0.500 + 32.28}\right)\left(\frac{10}{0.0069 + 10}\right)$$

$\therefore A_{vs} = 0.974$ Ans

vi) Overall current gain $(A_{is}) = \dfrac{V_L/R_L}{\dfrac{V_s}{r_s + r_{in(stage)}}}$

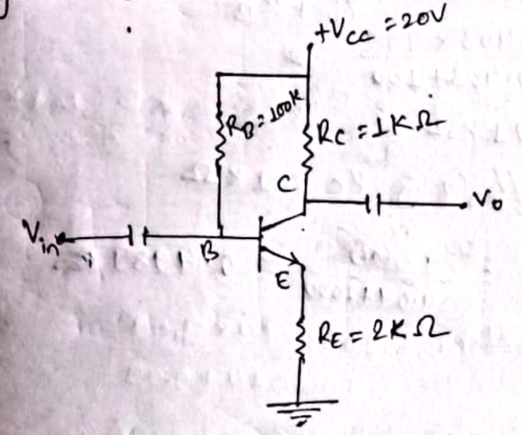
$$= \frac{V_L}{V_s} \times \left(\frac{r_s + r_{in(stage)}}{R_L}\right)$$

$$= 0.974 \times \left(\frac{0.5 + 32.28}{10}\right)$$

$\therefore A_{is} = 3.2$ Ans

2. Draw and find re model and find r_{in}, $r_{in(stage)}$, $r_{o(stage)}$, A_v, A_{vs} and A_{is}. (Take $\beta = 100$).

Soln The given circuit is:



Applying KVL in input loop, we get:-

$$V_{cc} = 100 I_B + V_{BE} + I_E R_E$$

$$\Rightarrow 20 = 100\frac{I_E}{101} + 0.7 + 2I_E$$

$$\Rightarrow 20 - 0.7 = 2.99$$

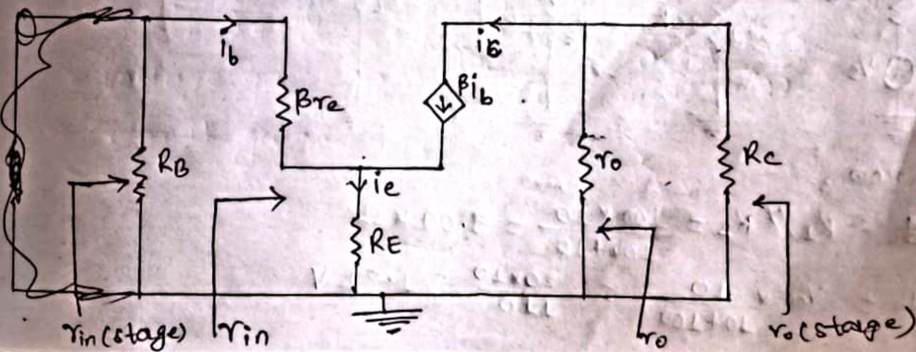
$$\Rightarrow \therefore \frac{19.3}{2.99} = I_E$$

$$\therefore I_E = 6.45\,mA$$

Now,

$$r_e = \frac{26\,mV}{6.45\,mA} = 4.03\,\Omega = 0.00403\,k\Omega$$

The re-model of given circuit is:



i) $r_{in} = \beta(r_e + R_E) = 100(0.00403 + 2) = 200.403\ K\Omega$

ii) $r_{in\,(stage)} = r_{in} // R_B = \frac{r_{in} \times R_B}{r_{in} + R_B}$

$= \frac{200.403 \times 100}{200.403 + 100}$

$= 66.71\ K\Omega$

iii) $r_{o\,(stage)} = R_C // r_o = R_C = 1K\Omega$

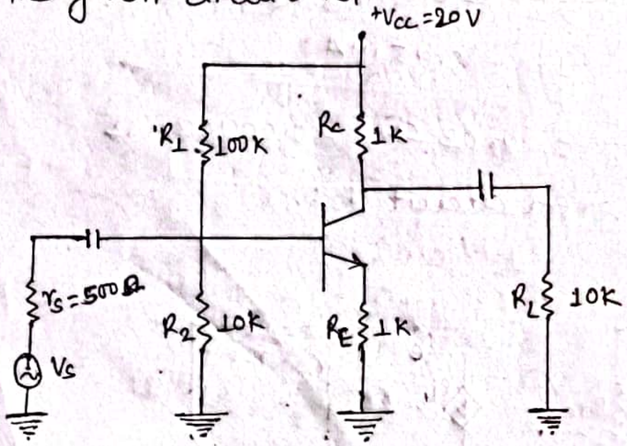
iv) $A_v = -\frac{R_C}{r_e + R_E} = \frac{1000}{4.09 + 2000} = 0.4989$

iv) $\cancel{A_{vs} = \frac{V_L}{V_s}}$

$= \cancel{A_v}\left(\frac{\cancel{r_{in(stage)}}}{r.}\right)$

Q.8. Draw r_e-model and find r_{in}, $r_{in\,(stage)}$, $r_{o(stage)}$, A_v, $\frac{V_L}{V_S}$ and $\frac{i_L}{i_L}$. Take $\beta = 100$

Sol The given circuit is:

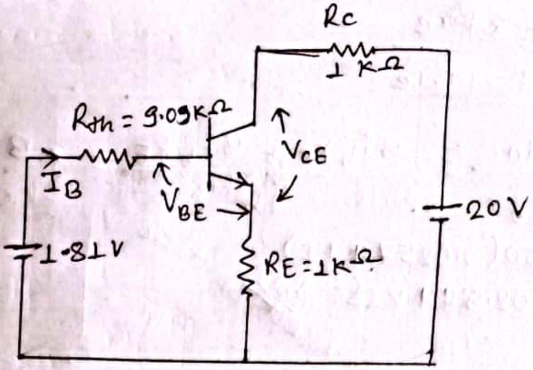


we know,

$R_B = R_{th} = R_1 // R_2 = \frac{100 \times 10}{100 + 10} = 9.09\ K\Omega$

$V_{th} = V_{cc} \times \frac{10}{10 + 100} = \frac{20 \times 10}{110} = 1.81\ V$

The Thevenin's equivalent circuit is:



Applying KVL in input loop, we get:

$1.81 = 9.09\, I_B + 0.7 + I_E$

$\Rightarrow 1.81 - 0.7 = 9.09 \times \frac{I_E}{100} + I_E$

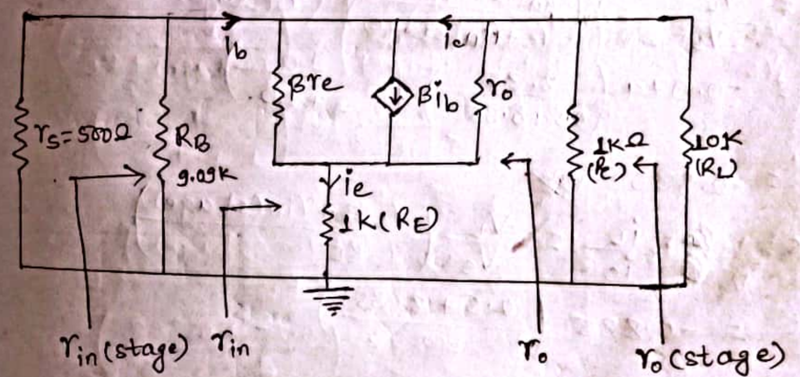
$\Rightarrow \frac{1.11}{1.0909} = I_E$

$\therefore I_E = 1.01\ mA$

Now,

$\therefore r_e = \frac{26mV}{1.01\ mA} = 25.74\ \Omega = 0.02574 K\Omega$

The r_e-model of given circuit is:



$\therefore \beta r_e =$

$\cancel{i> r_{in} = R_E // \beta r_e}$

$\cancel{R_E \times \beta r_e}$

$\cancel{R_E + \beta r_e}$

i> $r_{in} = \dfrac{V_{in}}{i_b} = \dfrac{\beta i_b r_e + \beta i_b R_E}{i_b} = \beta(r_e + R_E)$

$= 100(0.02574 + 1)$

$\therefore r_{in} = 102.574\ K\Omega$ Ans

ii> $r_{in}(stage) = r_{in} // R_B = \dfrac{102.574 \times 9.09}{102.574 + 9.09}$

$= 8.35\ K\Omega$ Ans

iii> $r_o(stage) = R_c // r_o = R_c \quad (\because r_o = \infty)$

$= 1\ K\Omega$ Ans

iv> $A_v = \cancel{\dfrac{R_c // r_o}{}} = \dfrac{-R_c}{r_e + R_E} = \dfrac{-1}{0.02574 + 1}$

$= -0.974$ Ans

v> $A_{vs} = \dfrac{V_L}{V_s} = A_v\left(\dfrac{r_{in}(stage)}{r_s + r_{in}(stage)}\right)\left(\dfrac{R_L}{r_o(stage) + R_L}\right)$

$= -0.974\left(\dfrac{8.35}{0.5 + 8.35}\right)\left(\dfrac{10}{1 + 10}\right)$

$= -0.835$ Ans

vi> $A_{is} = \dfrac{i_L}{i_s} = \dfrac{V_L}{V_s}\left(\dfrac{r_s + r_{in}(stage)}{R_L}\right)$

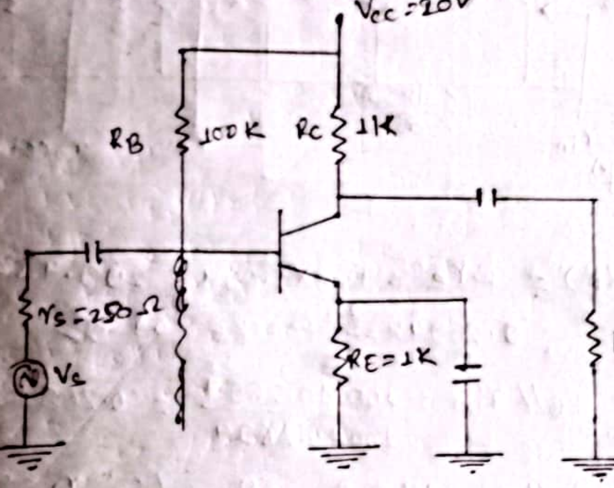
$= -0.835\left(\dfrac{0.5 + 8.35}{10}\right)$

$= -0.73$ Ans

Emitter bypassed

Q. For the transistor amplifier circuit shown below, find its r_e-model and r_{in}, $r_{in}(stage)$, $A r_o(stage)$, A_V, $\dfrac{V_L}{V_s}$ and also find the value of V_L if $V_s = 100$. Assume $\beta = 100$.

Sol The given amplifier circuit is:



Applying KVL in input loop, we get;

$V_{cc} = I_B R_B + V_{BE} + I_E R_E$

$\Rightarrow 20 = 100 I_B + 0.7 + I_E$

$\Rightarrow 20 - 0.7 = 100 \dfrac{I_E}{100} + I_E$

$\Rightarrow 19.3 = 2 I_E$

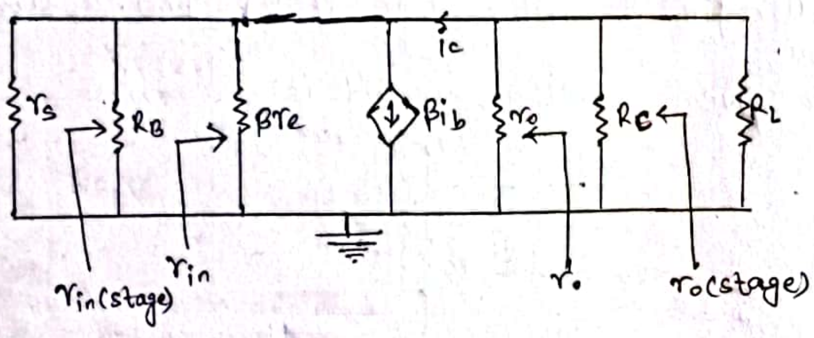
$\therefore I_E = 9.65 mA$

Now,

$r_e = \dfrac{26mV}{9.65mA} = 2.694\ \Omega$

$\therefore \beta r_e = 100 \times 2.694 = 0.2694\ k\Omega$

The r_e-model of given circuit is:-



i> $r_{in} = \beta(r_e + R_E) = \beta r_e = 100 \times 2.694 = 269.4\Omega$

$= 0.2694 k\Omega$ $(\because R_E = 0)$ Ans

ii> $r_{in(stage)} = R_B // r_{in} = \frac{100 \times 0.2694}{100 + 0.2694} = 0.268 k\Omega$ Ans

iii> $r_{o(stage)} = R_C // r_o = R_C = 1k\Omega$ Ans $(\because r_o = \infty)$

iv> $A_v = \frac{-R_C}{r_e + R_E} = \frac{-R_C}{r_e} = \frac{-1\times10^3}{2.694} = -371.2$ Ans

v> $A_{vs} = \frac{V_L}{V_s} = A_v \left(\frac{r_{in(stage)}}{r_s + r_{in(stage)}}\right) \times \left(\frac{R_L}{r_{o(stage)} + R_L}\right)$

$= -371.2 \left(\frac{0.268}{0.25 + 0.268}\right) \times \left(\frac{4}{1+4}\right)$

$= -153.639$ Ans

vi> $A_{is} = \frac{i_L}{i_s} = \frac{V_L}{V_s} \times \left(\frac{r_s + r_{in(stage)}}{R_L}\right)$

$= -153.639 \times \left(\frac{0.25 + 0.268}{4}\right)$

$= -19.896$ Ans

we know,

$V_s = 100$,

Now,

$\frac{V_L}{V_s} = -153.639$

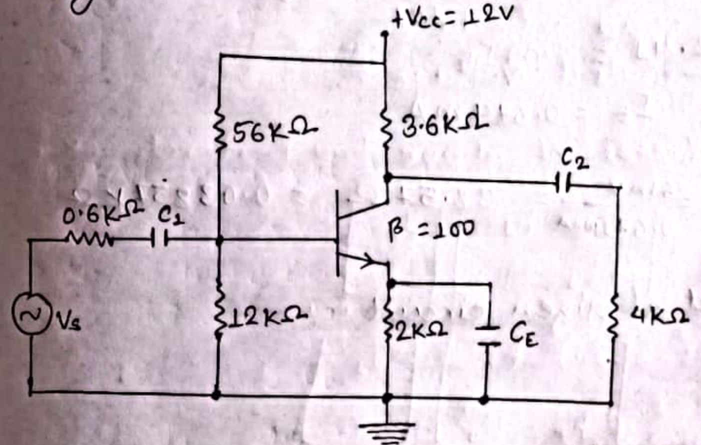
$V_L = -153.639 \times 100$ (V_s)

$\therefore V_L = -15363.9\ V$ Ans

Pre-Board - 2079

4.a. For the given common emitter amplifier. Find the following using re model. Input impedance, output impedance, voltage gains, $\beta = 100$, $V_{CC} = +12V$.

The give CE amplifier circuit is:-

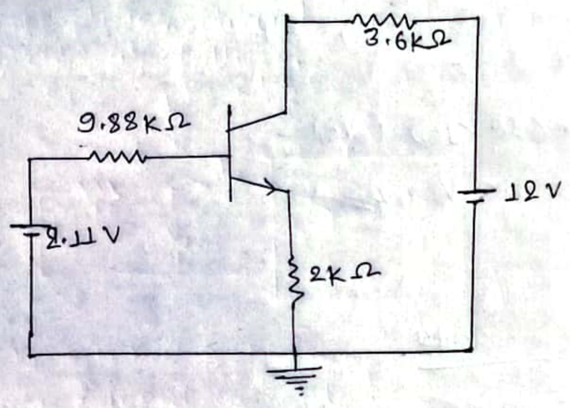


Here,

$R_B = R_{th} = R_1 // R_2 = \frac{56 \times 12}{56 + 12} = 9.88 k\Omega$

$V_{th} = V_{CC} \times \frac{R_2}{R_1 + R_2} = 12 \times \frac{12}{12 + 56} = 2.11\ V$

The thevenin's equivalent circuit is:



Applying KVL in input loop, we get:

$$2.11 = 9.88 I_B + 0.7 + 2 I_E$$

$$\Rightarrow 2.11 - 0.7 = 9.88 \times \frac{I_E}{101} + 2 I_E$$

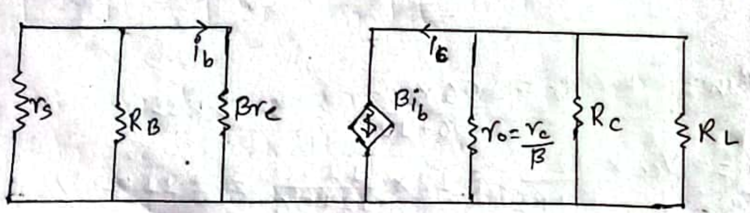
$$\Rightarrow 1.41 = 2.09 I_E$$

$$\therefore I_E = 0.674 \text{ mA}$$

Now,

$$r_e = \frac{26 mV}{0.674 mA} = 38.57 \Omega = 0.03857 K\Omega$$

The re model of given circuit is:



$$r_{in} = \beta r_e = 100 \times 0.03857 = 3.857 \text{ K}\Omega$$

i> Input impedance, $Z_{in} = r_{in} // R_B = \frac{r_{in} \times R_B}{r_{in} + R_B}$

$$= \frac{3.857 \times 9.88}{3.857 + 9.88}$$

$$= 2.774 \text{ K}\Omega \text{ Ans}$$

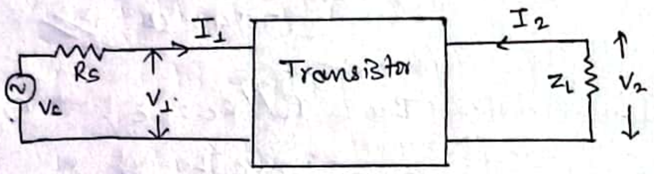
ii> output Impedance, $Z_o = r_o // R_C = R_C \quad (\because r_o = \infty)$

$$= 3.6 K\Omega \text{ Ans}$$

iii> $A_v = -\frac{R_C}{r_e} = \frac{-3.6}{0.03857} = -93.33$ Ans

Hybrid Parameters

A BJT amplifier circuit can be represented by two port network are as follows:



Here, V_1 = input voltage
I_1 = input current
R_s = source resistance
V_s = source voltage
I_2 = output current
Z_L = load impedance
V_2 = output voltage

The h-parameter of above two port network is:

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_i & h_r \\ h_f & h_o \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

where,
$h_{11} = h_i$ = input impedance (resistance)
$h_{12} = h_r$ = reverse voltage gain
$h_{21} = h_f$ = forward voltage gain
$h_{22} = h_o$ = output admittance

$$\Rightarrow V_1 = h_i I_1 + h_r V_2 \quad \text{——— (1)}$$
$$I_2 = h_f I_1 + h_o V_2 \quad \text{——— (2)}$$

From above two port network,
$$V_2 = -I_2 Z_L \quad \text{——— (3)}$$

The h-model of above two port network is:-

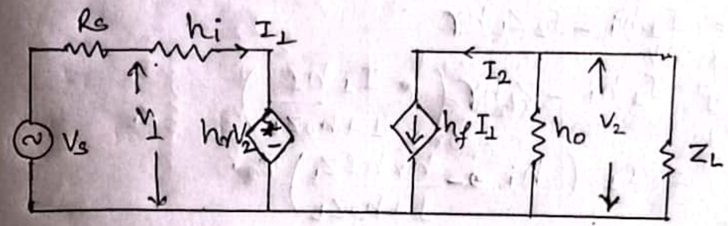


fig. h-model of two port network

i) Current gain, $A_i = -\frac{I_2}{I_1}$
From (2),
$$I_2 = h_f I_1 + h_o V_2$$
$$\Rightarrow I_2 = h_f I_1 + h_o(-I_2 Z_L) \quad (\because \text{from (3)})$$
$$\Rightarrow I_2 + I_2 h_o Z_L = h_f I_1$$
$$\Rightarrow \frac{I_2}{I_1} = \frac{h_f}{1 + h_o Z_L}$$
$$\therefore A_i = -\frac{I_2}{I_1} = -\frac{h_f}{1 + h_o Z_L}$$

ii) Voltage gain, $A_v = \frac{V_2}{V_1} = \frac{-I_2 Z_L}{V_1}$
$$= \frac{-A_i I_1 Z_L}{V_1}$$
$$= \frac{-A_i Z_L}{V_1 / I_1} = \frac{-A_i Z_L}{Z_i}$$
$$\therefore A_v = \frac{-A_i Z_L}{Z_i}$$

iii) Input impedance, $Z_i = \frac{V_1}{I_1}$

from (1),

$V_1 = I_1 h_i + h_r V_2 = I_1 h_i + h_r(-I_2 Z_L)$

$\Rightarrow V_1 = I_1 h_i - I_2 Z_L h_r$

$\Rightarrow V_1 = I_1 h_i - Z_L h_r \left(\frac{-I_1 h_f}{1+h_o Z_L} \right)$

$\Rightarrow V_1 = I_1 \left(h_i - \frac{h_r h_f Z_L}{1+h_o Z_L} \right)$

$\Rightarrow \frac{V_1}{I_1} = h_i - \frac{h_r h_f}{\frac{1}{Z_L} + h_o}$

$\therefore \frac{V_1}{I_1} = h_i - \frac{h_r h_f}{Y_L + h_o}$

iv) output admittance, $Y_o = \frac{I_2}{V_2}$

From (2),

$I_2 = h_f I_1 + h_o V_2$

$\Rightarrow I_2 = \frac{h_f I_2}{\frac{h_f}{1+h_o Z_L}} + h_o V_2$

$\Rightarrow I_2 = I_2(1+h_o Z_L) + h_o V_2$

$\Rightarrow (1-1-h_o Z_L) I_2 = h_o V_2$

$\Rightarrow \frac{I_2}{V_2} = \frac{h_o}{-h_o Z_L} = -\frac{1}{Z_L}$

$\therefore Y_o = -\frac{1}{Z_L}$

~~Again,~~

from (2), $I_2 = h_f I_1 + h_o V_2$

$\Rightarrow \frac{I_2}{V_2} = h_f \frac{I_1}{V_2} + h_o$

$\Rightarrow Y_o = h_o + h_f \frac{I_1}{V_2}$ —— (a)

from (2) & (3), we get:

$\frac{-V_2}{Z_L} = h_f I_1 + h_o V_2$

$\Rightarrow$

Applying KVL in input loop, we get:

$V_s = I_1 R_s + h_i I_1 + h_r V_2$

$\Rightarrow 0 = I_1(h_i + R_s) + h_r V_2$

$\Rightarrow -h_r V_2 = I_1(h_i + R_s)$

$\therefore \frac{I_1}{V_2} = \frac{-h_r}{h_i + R_s}$

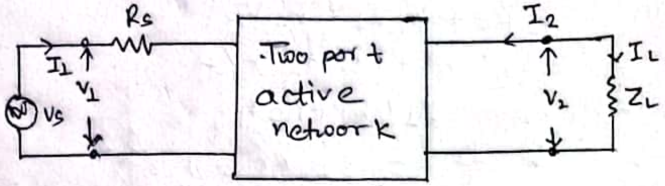
From (a),

$Y_o = h_o - \frac{h_r h_f}{h_i + R_s}$

$\therefore Z_o = \frac{1}{Y_o}$

h- parameter or h-model parameter

A BJT amplifier circuit can be represented by two port network as follows;



Here, I_1 = input current
V_1 = input voltage
R_s = source resistance
V_s = ~~volt~~ source voltage
I_2 = output voltage
V_2 = output voltage
I_L = Load current
Z_L = Load impedance

The h-parameter of above two port network is:-

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_i & h_r \\ h_f & h_o \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

where, $h_{11} = h_i$ = input impedance (resistance)
$h_{12} = h_r$ = reverse voltage gain
$h_{21} = h_f$ = forward voltage gain
$h_{22} = h_o$ = output admittance

$\Rightarrow$ $V_1 = h_i I_1 + h_r V_2$ ——— (1)
$I_2 = h_f I_1 + h_o V_2$ ——— (2)

From output loop of two port network,
$V_2 = - I_2 Z_L$ ——— (3)

~~From input loop of two port network,~~

The h-model of two port network is:-

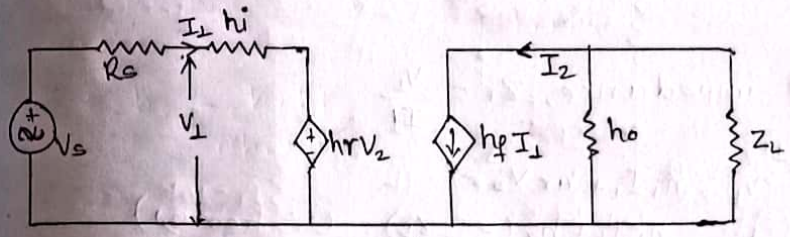


fig. h- model of two portnetwork

~~i) Input impedance, $Z_i = \frac{V_1}{I_1}$~~

i) Current gain, $A_i = -\frac{I_2}{I_1}$

From (2),

~~$V_1 = h_i I_1 + h_r V_2$~~

$\Rightarrow$ ~~$V_1 = h_i I_1 - I_2 Z_L h_o$~~

$\cdot$ $I_2 = h_f I_1 + h_o V_2$

$\Rightarrow I_2 = h_f I_1 - I_2 Z_L h_o$

$\Rightarrow I_2 + I_2 Z_L h_o = h_f I_1$

$\Rightarrow I_2 (1 + Z_L h_o) = h_f I_1$

$\Rightarrow \frac{I_2}{I_1} = \frac{h_f}{1 + Z_L h_o}$

$\therefore A_i = -\frac{I_2}{I_1} = \frac{-h_f}{1 + Z_L h_o}$

ii) Voltage gain, $A_v = \frac{-V_2}{V_1} = \frac{-I_2 Z_L}{V_1}$ (∵ from (3))

$= \frac{-I_2 Z_L}{V_1} = -\frac{Z_L \times I_1 A_i}{V_1}$

$= \frac{-A_i Z_L}{V_1/I_1}$

$\therefore A_v = -\frac{A_i Z_L}{Z_i}$

iii) Input impedance, $Z_i = \frac{V_1}{I_1}$

From (1),

$V_1 = h_i I_1 + h_r V_2$

$\Rightarrow V_1 = h_i I_1 + h_r(-I_2 Z_L)$ (∵ from (3))

$\Rightarrow V_1 = h_i I_1 - h_r Z_L I_2$

$\Rightarrow V_1 = h_i I_1 - h_r Z_L \left(\frac{h_f}{1+Z_L h_o}\right)$

$\Rightarrow V_1 = I_1 \left(h_i - \frac{h_r h_f}{\frac{1}{Z_L} + h_o}\right)$

$\therefore \frac{V_1}{I_1} = h_i - \frac{h_r h_f}{h_o + Y_L}$

$\therefore Z_i = h_i - \frac{h_r h_f}{h_o + Y_L}$

iv) Output admittance (Y_o) and ~~Input~~ Output impedance, Z_o:

$Z_o = \frac{I_2}{V_2}$

From (2),

$I_2 = h_f I_1 + h_o V_2$

$\Rightarrow \frac{I_2}{V_2} = h_f \frac{I_1}{V_2} + h_o$

$\Rightarrow Z_o = h_o + h_f \frac{I_1}{V_2}$ —— (a)

Applying KVL in input loop of h-model, we get:

$V_s = I_1 R_s + I_1 h_i + V_2 h_r$

$\Rightarrow 0 = I_1 (R_s + h_i) + V_2 h_r$

$\Rightarrow -V_2 h_r = I_1 (R_s + h_i)$

$\therefore \frac{I_1}{V_2} = -\frac{h_r h_f}{R_s + h_i}$

From (a),

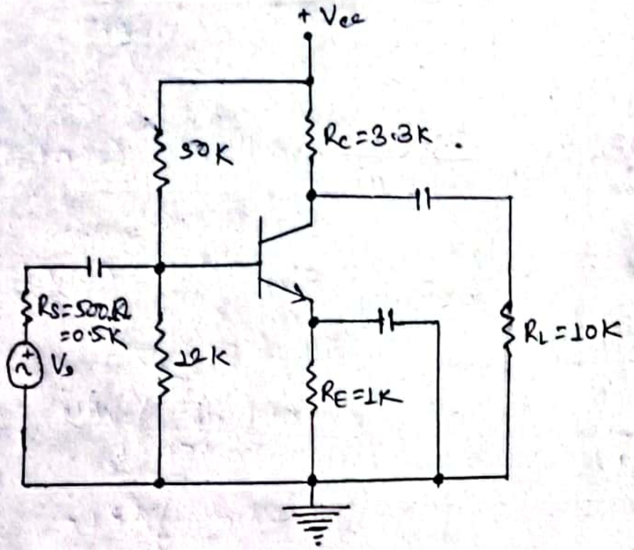
$Y_o = h_o - \frac{h_r h_f}{R_s + h_i}$

$\therefore Z_o = \frac{1}{Y_o}$

Numericals:

1. For the given transistor amplifier circuit the h-parameters are $h_{ie} = 1600\Omega$, $h_{fe} = 80$, $h_{re} = 20\times10^{-4}$, and $h_{oe} = 20\mu$ Find Z_i, Z_i(stage), Z_o, Z_o(stage), A_i, A_v, A_{vs}.

Sol: The given amplifier circuit is:-



Given:-

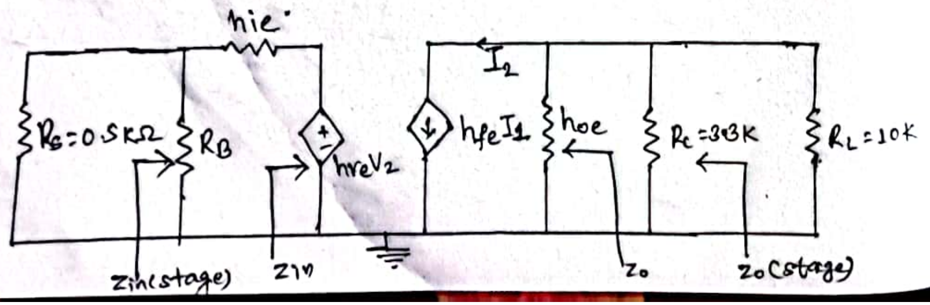
$h_{ie} = 1600\,\Omega = 1.6\,k\Omega$

$h_{fe} = 80$

$h_{re} = 20\times10^{-4}$

$h_{oe} = 20\times10^{-6}\,S$

The h- model of given two port network is:



$V_1 = h_{ie} I_1 + h_{re} V_2$

$I_2 = h_{fe} I_1 + h_{oe} V_2$

we know,

$$Z_L = R_C // R_L = \frac{R_C \times R_L}{R_C + R_L} = \frac{3.3\times10}{3.3+10} = 2.48\,k\Omega$$

$$R_B = R_1 // R_2 = \frac{50\times12}{50+12} = 9.67\,k\Omega$$

i) Input impedance, $Z_{in} = h_{ie} - \dfrac{h_{fe}.h_{re}}{h_{oe} + Y_L}$

$$= 1.6 - \frac{80\times20\times10^{-4}}{20\times10^{-6} + \frac{1}{2.48}}$$

$\therefore Z_{in} = 1.2\,k\Omega$

ii) Output impedance, $Z_o = \dfrac{1}{Y_o}$

we know,

$$Y_o = h_{oe} - \frac{h_{fe}.h_{re}}{h_{ie} + (R_S // R_B)}$$

$$\Rightarrow Y_o = 20\times10^{-6} - \frac{80\times20\times10^{-4}}{\left(1.6 + \frac{0.5\times9.67}{0.5+9.67}\right)\times10^{3}}$$

~~$\Rightarrow Y_o = 20\times10^{-6} - 0.77\times10^{-6}$~~

$\Rightarrow Y_o = -5.7\times10^{-5}$

$\therefore Y_o = 5.7\times10^{-5}$ (neglecting negative sign)

Now,

$$Z_o = \frac{1}{5.7\times10^{-5}} = 17.543\,k\Omega$$

iii> $A_{i} = \frac{-h_{fe}}{1+h_{oe}Z_L} = \frac{-80}{1+20\times10^{-6}\times2.48\times10^{3}}$

$= -76.22$ Ans

iv> $A_v = \frac{-A_i Z_L}{Z_{in}} = \frac{-76.22\times2.48}{1.2} = -157.52$ Ans

v> $Z_{in}(stage) = Z_{in} // R_B = \frac{1.2\times9.67}{1.2+9.67} = 1.067 K\Omega$ Ans

vi> $Z_o(stage) = Z_o // R_C = \frac{17.543\times3.3}{17.543+3.3}$

$= 2.77 K\Omega$ Ans

vii> $A_{vs} = A_v\left(\frac{Z_{in}(stage)}{R_s+Z_{in}(stage)}\right)\times\left(\frac{R_L}{R_L+Z_o(stage)}\right)$

$= -157.52\left(\frac{1.067}{0.5+1.067}\right)\times\left(\frac{10}{10+2.77}\right)$

$\therefore A_{vs} = -83.99$ Ans

viii> $A_{is} = A_{vs}\left(\frac{R_s+Z_{in}(stage)}{R_L}\right)$

$= -83.99\left(\frac{0.5+1.067}{10}\right)$

$\therefore A_{is} = 13.16$ Ans

1. The transistor shown in figure below has following h-parameters are $h_{ie} = 1K\Omega$, $h_{re} = 2.5\times10^{-4}$, $h_{fe} = 50$, $h_{oe} = 25\mu S$. Find

i> input impedance, Z_{in}

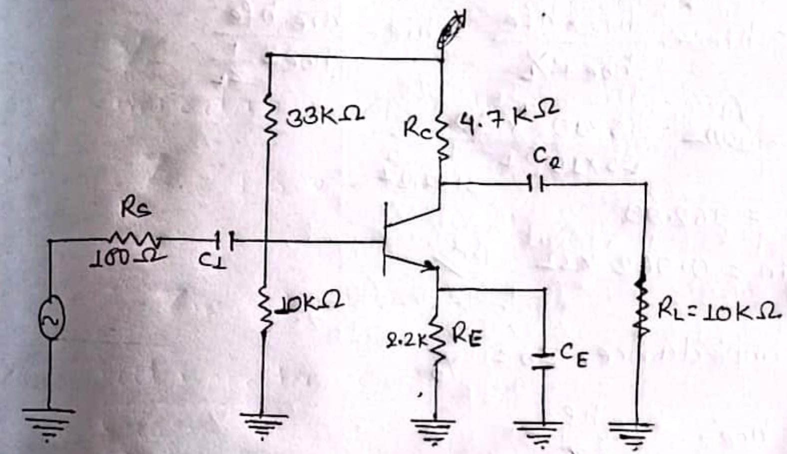
ii> output impedance, Z_{out}

iii> voltage gain, A_v

iv> current gain, A_i

(2013-Fall)

Sol The given transistor is:



Given:-

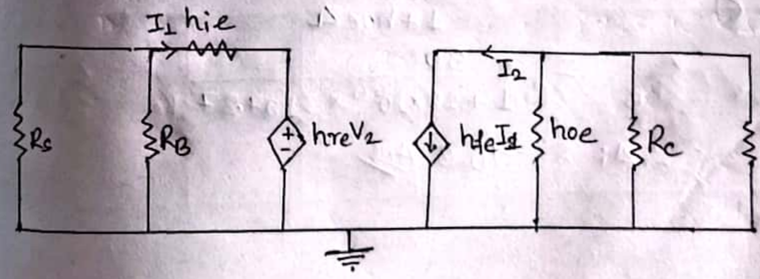
$h_{ie} = 1K\Omega$

$h_{re} = 2.5\times10^{-4}$

$h_{fe} = 50$

$h_{oe} = 25\mu S = 25\times10^{-6} S$

The h-model of given transistor is:



Now,

$R_B = R_1 // R_2 = \frac{10 \times 33}{10+33} = 7.674 K\Omega = 7674.4\Omega$

$Z_L = R_C // R_L = \frac{4.7 \times 10}{4.7+10} = 3.197 K\Omega = 3197.27\Omega$

$Z_S = R_S // R_B = \frac{100 \times 7674.4}{100 + 7674.4} = 98.71 \Omega$

i) Input impedance, Zin:

$Z_{in} = h_{ie} - \frac{h_{re} \times h_{fe}}{h_{oe} + Y_L} = h_{ie} - \frac{h_{re} h_{fe}}{h_{oe} + \frac{1}{Z_L}}$

$= 1000 - \frac{2.5 \times 10^{-4} \times 50}{25 \times 10^{-6} + \frac{1}{3197.27}}$

$= 962.99$

$\therefore Z_{in} = 0.962 K\Omega$ Ans

ii) output impedance, $Z_o = \frac{1}{Y_o}$

Now,

$Y_o = h_{oe} - \frac{h_{re} h_{fe}}{h_{ie} + Z_s}$

$= 25 \times 10^{-6} - \frac{2.5 \times 10^{-4} \times 50}{1000 + 98.71}$

$\therefore Y_o = 1.36 \times 10^{-5}$

$\therefore Z_o = \frac{1}{Y_o} = \frac{1}{1.36 \times 10^{-5}} = 7.33 \times 10^4 = 73.33 K\Omega$ Ans

~~iii) Voltage gain,~~

iv) Current gain, $A_i = -\frac{h_{fe}}{1 + h_{oe} Z_L}$

$= -\frac{50}{1 + 25 \times 10^{-6} \times 3197.27}$

$\therefore A_i = -46.3$ Ans

iii) Voltage gain, $A_v = \frac{A_i Z_L}{Z_{in}}$

$= \frac{-46.3 \times 3197.27}{962.99}$

$\therefore A_v = -153.72$ Ans

v) Voltage gain with Rs, Avs: Avs:

$A_{vs} = A_v \times \frac{Z_{in}}{Z_{in} + Z_s}$

$= -153.72 \times \frac{962.99}{962.99 + 98.71}$

$\therefore A_{vs} = -139.42$ Ans

2014. Fall

2. For the CE amplifier, find the following using re-model

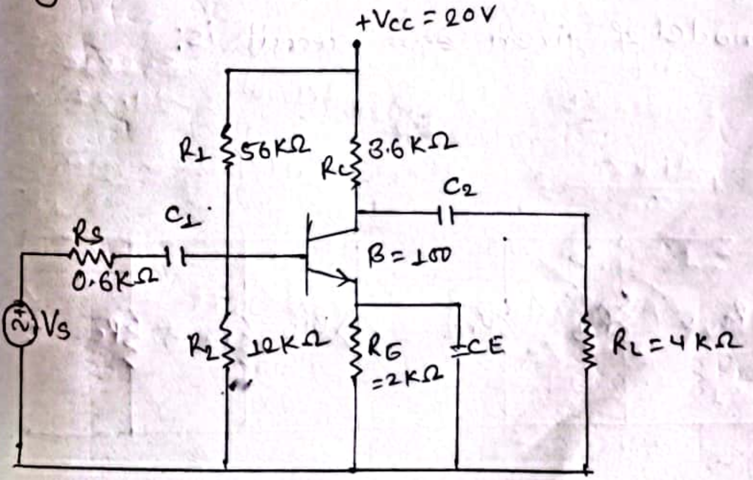
i) input impedance, Zin

ii) output impedance, Zo

iii) Curren gain, Ai

iv) Voltage gain, Av (Assume $\beta = 100$, $V_{cc} = +20V$)

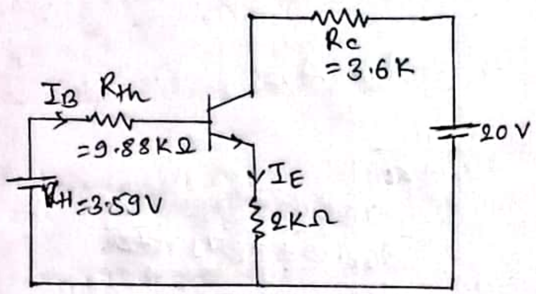
Sol: The given CE amplifier circuit is:



$R_B = R_{th} = R_1 // R_2 = \frac{R_1 \times R_2}{R_1+R_2} = \frac{12\times56}{12+56} = 9.88\,K\Omega$

$V_{th} = V_{cc} \times \frac{12}{12+56} = 20\times\frac{12}{12+56} = 3.59\,V$

The Thevenin's equivalent circuit is:-



Applying KVL in input loop, we get;

$3.59 = 9.88 I_B + 0.7 + 2I_E$

$\Rightarrow 3.59-0.7 = 9.88\frac{I_E}{100} + 2I_E$

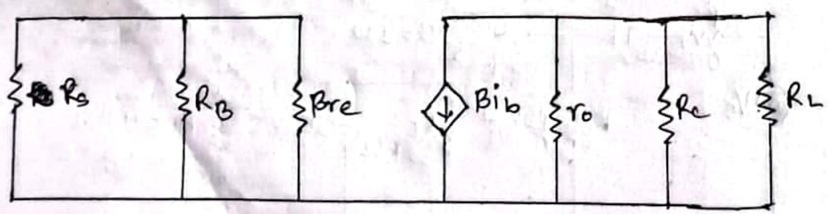
$\Rightarrow 2.89 = 2.09\, I_E$

$\therefore I_E = 1.38\, mA$

Now,

$r_e = \frac{26mV}{1.38mA} = 18.84\,\Omega = 0.01884\,K\Omega$

The re model of given ~~equ~~ circuit is:



i) Input impedance, $Z_{in} = R_B // \beta r_e = \frac{100\times0.01884\times9.88}{100\times0.01884+9.88}$

$\therefore Z_{in} =$ ~~1.884 KΩ~~ $1.848\,K\Omega$

ii) Output impedance, $Z_o = r_o // R_c // R_L = R_c // R_L$

$= 3.6 // 4$

$= \frac{3.6\times4}{3.6+4}$

$\therefore Z_o = 1.894\,K\Omega$

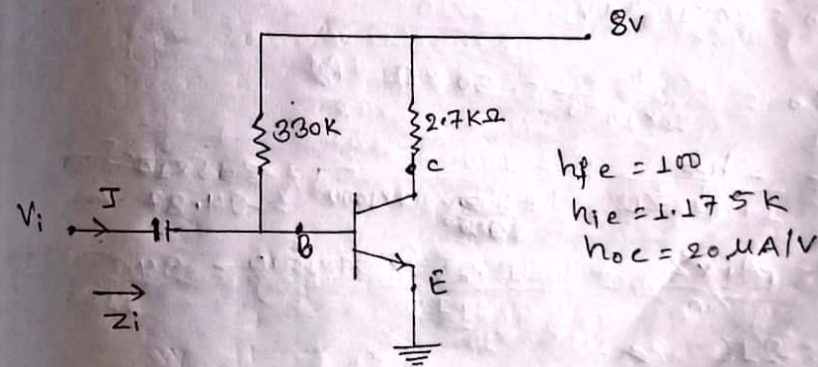
iii) Voltage gain, $A_v = -\frac{R_c}{r_e} = \frac{-3.6}{0.01884} = -191.08$ Ans

iv) Current gain, $A_I = \beta = -100$ Ans

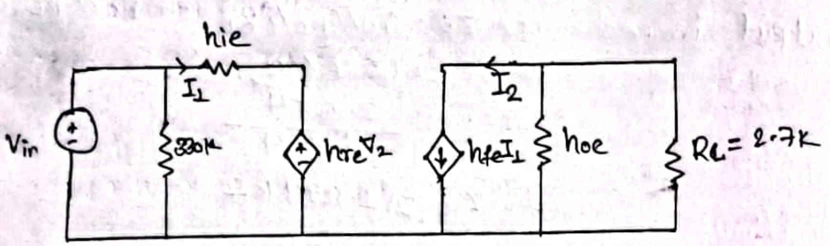
2015-Fall

Q. Find A_{vs}, A_v, Z_o, Z_{in} of the following circuit Diagram using h-parameters
$h_{fe} = 100$, $h_{ie} = 1.175\,K\Omega$, $h_{oe} = 20\,\mu A/V$.

Ans: The given circuit diagram is;



The h-model of given circuit is',



i) $Z_{in} = hie - \frac{hfe \times hre}{hoe + Z_L} = hie - \frac{hfe \times hre}{hoe + \frac{1}{Y_L}}$

$= 1.175 \times 1000 - \frac{0}{20 \times 10^{-6} + \frac{1}{R_c}}$

$\therefore Z_{in} = 1.175\,k\Omega$ Ans

ii) $Z_o = \frac{1}{Y_o}$

Now,

$Y_o = hoe - \frac{hre \times hfe}{Z_s + hie} = 20 \times 10^{-6}\,A/V - 0$

$\therefore Y_o = 20 \times 10^{-6}\,AV^{-1}$

$\therefore Z_o = \frac{1}{20 \times 10^{-6}} = 50\,k\Omega$ Ans

iii) $A_i = \frac{-hfe}{1 + hoe \times Z_L} = \frac{-100}{1 + 20 \times 10^{-6} \times 2.7 \times 1000} = -94.87$ Ans

iv) $A_v = \frac{-A_i \times Z_L}{Z_{in}} = \frac{-94.87 \times 2.7 \times 1000}{1175}$

$= -218$ Ans

v) $A_{vs} = A_v \times \left(\frac{Z_{in}}{Z_{in} + Z_s}\right) = -218 \times \frac{1175}{1175 + 0}$

$= -218$ Ans

2021-Spring

Q.6. For the network given, determine the following parameters using the complete hybrid equivalent model:

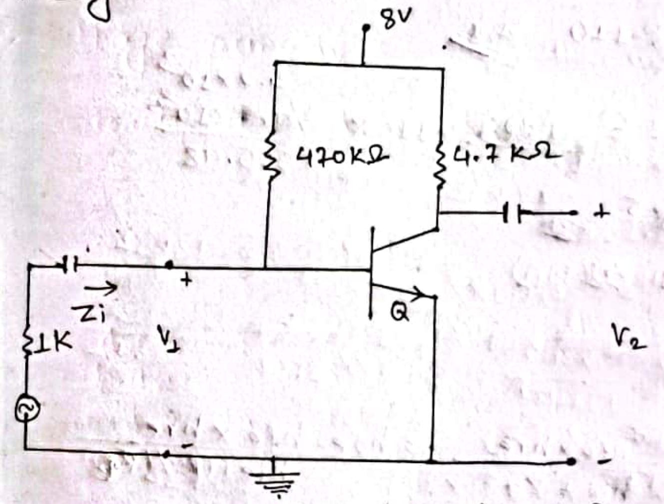
i) Z_i and Z_i'

ii) A_v

iii) $A_i = I_o/I_i$

iv) Z_o & Z_o'

Sol The given network is:-



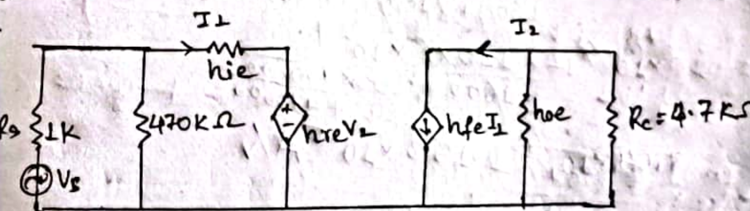
$h_{fe} = 110$

$h_{ie} = 1.6\,K\Omega = 1600\,\Omega$

$h_{re} = 2 \times 10^{-4}$

$h_{oe} = 20\,\mu AV^{-1} = 20 \times 10^{-6}\,AV^{-1}$

The hybrid model of given network is:



i) Input impedance, $Z_{in} = hie - \frac{hfe.hre}{hoe + Z_L} = hie - \frac{hfe.hre}{hoe + \frac{1}{Y_L}}$

$= 1600 - \frac{110 \times 2 \times 10^{-4}}{20 \times 10^{-6} + \frac{1}{4.7 \times 10^3}}$

$\therefore Z_{in} = 1505.48\,\Omega$ Ans

$Zin = R_{ss} || zin$

$= \frac{1000\times 1505.48}{1000+1505.48}$

$\therefore Zin = 600.874\ \Omega$ Ans

ii) $A_i = -\frac{h_{fe}}{1+h_{oe}\times Z_L} = \frac{-110}{1+20\times10^{-6}\times\frac{1}{4.7\times10^{3}}}$

$\therefore A_i = -109.99 \approx -110$ Ans

iii) $A_v = A_i \times \frac{Z_L}{Z_{in}} = -110 \times \frac{4.7\times10^{3}}{1505.48}$

$= \cancel{1.55\times10^{5}}$

$= -343.41$ Ans

iv) $Z_o = \frac{1}{Y_o}$

we know,

$Y_o = h_{oe} - \frac{h_{fe}\times h_{re}}{h_{ie}+Z_s} = h_{oe} - \frac{h_{fe}\times h_{re}}{h_{ie}+(R_s'//R_B)}$

$= 20\times10^{-6} - \frac{110\times2\times10^{-4}}{1600+(1000//47000)}$

$= 20\times10^{-6} - \frac{110\times2\times10^{-4}}{1600+\frac{1000\times470000}{1000+470000}}$

$\therefore Y_o = 1.153\times10^{-5}\ A/V$

Now,

$Z_o = \frac{1}{Y_o} = \frac{1}{1.153\times10^{-5}} = 86.71\ k\Omega$

Also,

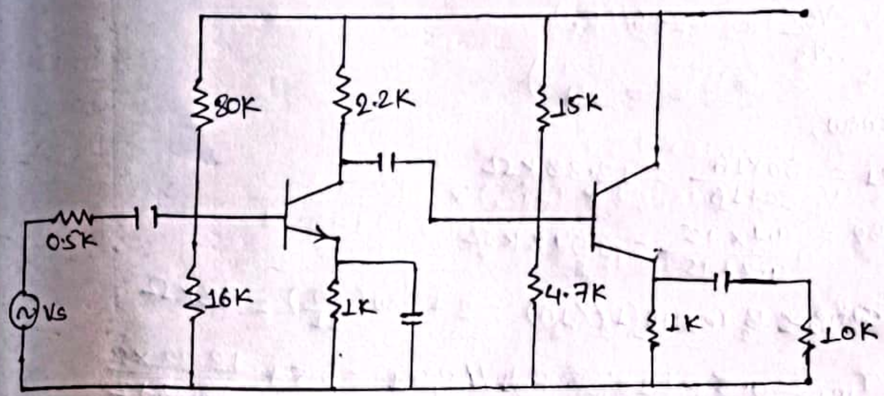
$Z_o' = Z_o // Z_L = \frac{86.71\times4.7}{86.71+4.7}$

$= 4.45\ k\Omega$ Ans

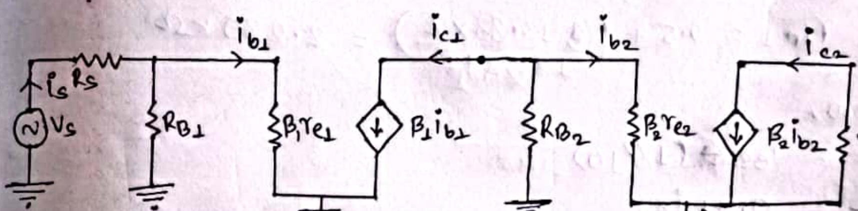
2019-Spring

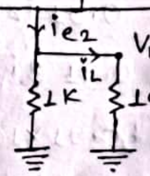
4.b Find V_L/V_s in the circuit below. ($\beta_1=\beta_2=100$, $r_{e1}=r_{e2}=20\Omega$)

Soln The given circuit is:



The re-model of given circuit is:





$R_{B1} = \frac{80\times16}{80+16} = 13.33k$

$R_{B2} = \frac{15\times4.7}{15+4.7} = 3.578\ K$

Now,

$V_L = i_{e2}(1//10) = \frac{10}{11} i_{e2} = 0.091\, i_{e2}$

$\Rightarrow V_L = 0.91\, i_{e2} = \beta i_{b2}(0.91)$

$\Rightarrow V_L = 100\times0.91\, i_{b2}$

$\Rightarrow V_L = 91 \times \frac{R_{B2}}{\beta_2 r_{e2}\times R_{B2}} \times (-i_{c1})$

$\Rightarrow V_L = -91\, i_{c1} \times \frac{3.578\times1000}{100\times20+3.578\times1000} = -58.37\, i_{c1}$

→ $V_L = -58.37\, i_{c1}$

→ $V_L = -58.37 \times \beta_1 i_{b1}$

→ $V_L = -58.37 \times 100 \times i_s \times \dfrac{13.33 \times 1000}{13.33 \times 1000 + 2000}$

→ $V_L = -7678.68\, i_s$

→ $V_L = -7678.38 \times \dfrac{V_s}{r_{in}}$

→ $\dfrac{V_L}{V_s} = -76$

We know,

$R_{B1} = \dfrac{80 \times 16}{80+16} = 13.33\,k\Omega$

$R_{B2} = \dfrac{4.7 \times 15}{4.7+15} = 3.578\,k\Omega$

$R_{b2} = \beta_2 r_{e2} + \beta_2 (1//10) = 2 + 100\left(\dfrac{10}{11}\right) = 93\,k\Omega$

~~$r_{in1} = 0.5 + (13.33\ R_{B1} // \beta_1 r_{e1}) = 0.5 + \dfrac{13.33 \times 2}{13.33+2}$~~

$R_{b1} = \beta_1 r_{e2} = 2\,k\Omega$

$r_{in1} = 0.5 + \left(\dfrac{13.33 \times 2}{13.33+2}\right) = 2.239\,k\Omega$

We know,

$V_L = i_{e2} (1//10)$

→ $V_L = 0.91\, i_{e2}$

→ $V_L = 0.91\, \beta i_{b2}$

→ $V_L = 0.91 \times 100 \times (-i_{c1}) \times \dfrac{3.578}{3.578+93}$

→ $V_L = -3.371\, i_{c1}$

→ $V_L = -3.371 \times \beta_1 i_{b1}$

→ $V_L = -3.371 \times 100 \times i_{b1}$

→ $V_L = -337.1 \times i_s \times \dfrac{13.33}{13.33+2}$

→ $V_L = -293\, i_s$

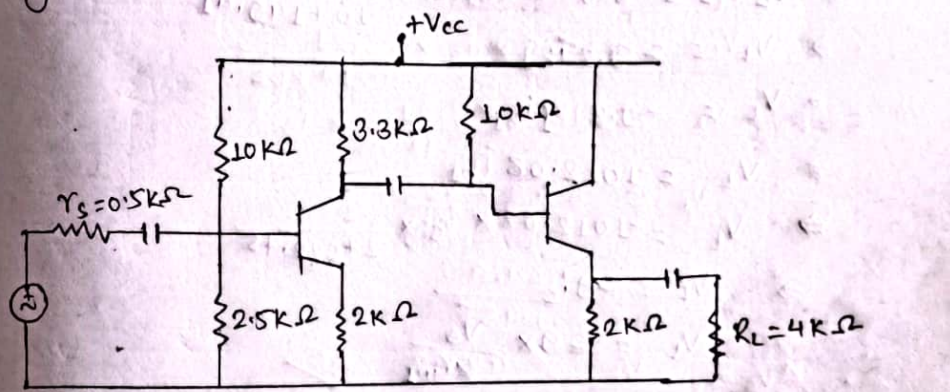
→ $V_L = -293.1 \times \dfrac{V_s}{R_{in1}}$

→ $\dfrac{V_L}{V_s} = \dfrac{-293.1}{2.239} = -130.9$ Ans

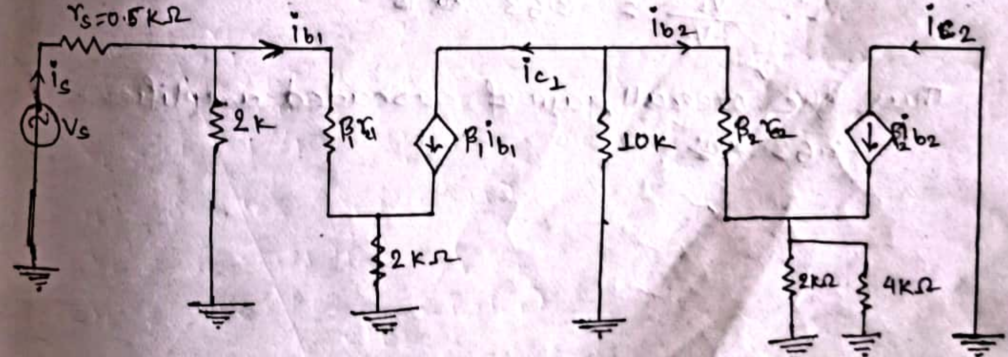
2015-Spring

Q. Find the overall voltage gain of cascaded transistor amplifier as shown in figure below. (given = $r_{e1} = r_{e2} = 25\,\Omega$ and $\beta_1 = \beta_2 = 110$).

Sol The given cascaded amplifier is:



The re model of given cascaded amplifier is:



$R_{b2} = \beta_2(r_e + R_{E2}//R_L) = 110\left(25\times10^{-3} + \frac{2\times4}{2+4}\right) = 149.41\,K\Omega$

$R_{b1} = \beta_1(r_e + R_{E1}) = 110(25\times10^{-3}+2) = 222.75\,K\Omega$

$r_{in1} = r_s + (2//R_{b1}) = 0.5 + \left(\frac{2\times222.75}{2+222.75}\right) = 2.48\,K\Omega$

we know that,

$V_L = i_{e2}(R_{E2}//R_L)$

$\Rightarrow V_L = \beta_2 i_{b2}\left(\frac{2\times4}{2+4}\right)$

$\Rightarrow V_L = 110\times\frac{4}{3}\times i_{b2}$

$\Rightarrow V_L = 146.67\times(-i_{c1})\times\frac{10}{10+149.41}$

$\Rightarrow V_L = -9.2\, i_{c1}$

$\Rightarrow V_L = -9.2\,\beta_1 i_{b1}$

$\Rightarrow V_L = -1012.08\, i_{b1}$

$\Rightarrow V_L = -1012.08\times i_s\times\frac{2}{2+222.75}$

$\Rightarrow V_L = -9 i_s$

$\Rightarrow V_L = -9\times\frac{V_s}{r_{in1}}$

$\Rightarrow \frac{V_L}{V_s} = \frac{-9}{2.48}$

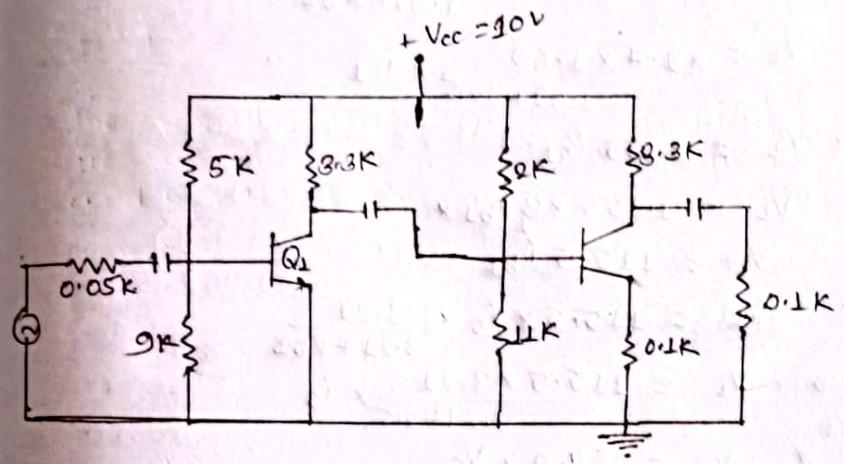
$\therefore \frac{V_L}{V_s} = -3.63$

Thus, the overall gain of cascaded amplifier is -3.63 Ans

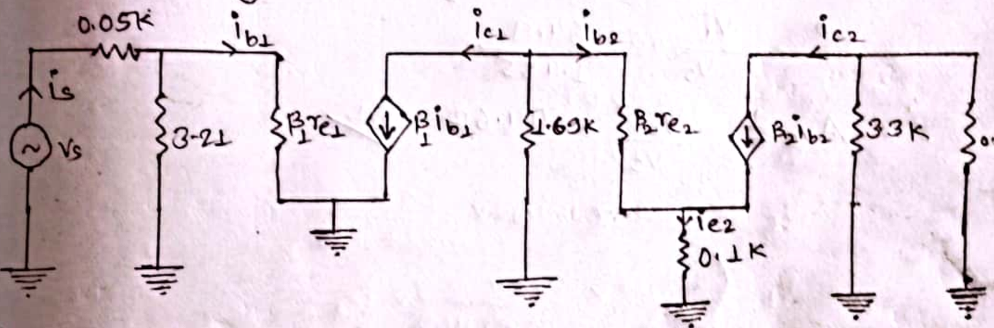
2019- fall

4.b. Find $\frac{V_L}{V_s}$ in the circuit below. ($\beta_1 = \beta_2 = 100$, $r_{e1} = r_{e2} = 25\Omega$)

Sol: The given circuit is.



The re-model of given circuit is:-



we have,

$R_{b2} = \beta_2(r_{e2}+0.1) = 100(0.025+0.1) = 12.5\,K\Omega$

$R_{b1} = \beta_1 r_{e1} = 100\times0.025 = 2.5\,K\Omega$

$r_{in1} = 0.5 + (3.21//R_{b1}) = 0.05 + \left(\frac{3.21\times2.5}{3.21+2.5}\right) = 1.455\,K\Omega$

we know that,

$V_L = -i_{c2}\times(3.33//0.1)$

$\Rightarrow V_L = -i_{c2}\times\frac{3.33\times0.1}{3.33+0.1}$

$\Rightarrow V_L = -0.097\, i_{c2}$

$\Rightarrow V_L = -0.097\ \beta_2 i_{b2}$

$\Rightarrow V_L = -0.097 \times 100\ i_{b2}$

$\Rightarrow V_L = -9.7 i_{b2}$

$\Rightarrow V_L = -9.7 \times (-i_{c1}) \times \frac{1.69}{1.69 + R_{b2}}$

$\Rightarrow V_L = 9.7 \times \frac{1.69}{1.69+125} \times i_{c1}$

$\Rightarrow V_L = 1.155\ i_{c1}$

$\Rightarrow V_L = 1.155 \times \beta_1 i_{b1}$

$\Rightarrow V_L = 115.5\ i_{b1}$

$\Rightarrow V_L = 115.5 \times i_s \times \frac{3.21}{3.21 + R_{b2}}$

$\Rightarrow V_L = \frac{115.5 \times 3.21}{3.21+2.5} \times i_s$

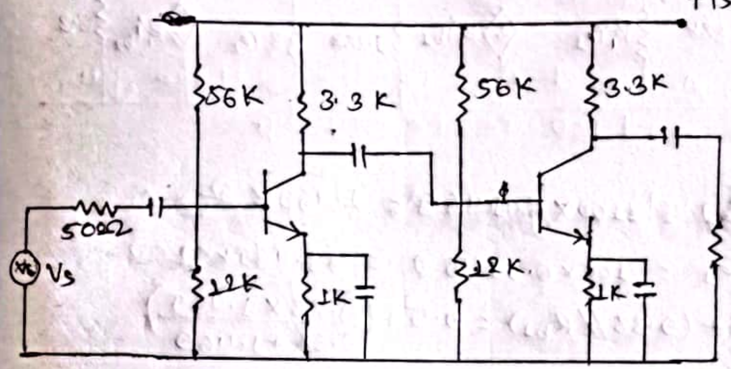
$\Rightarrow V_L = 64.9 \times \frac{V_s}{r_{in1}}$

$\Rightarrow \frac{V_L}{V_s} = \frac{64.9}{1.455}$

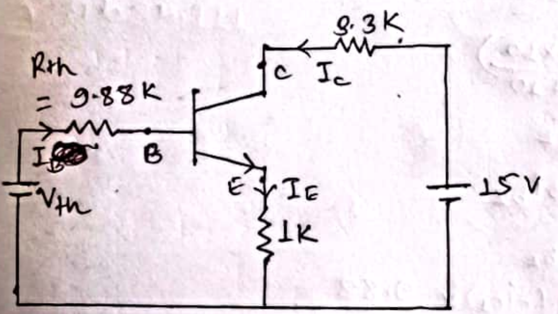
$\therefore A_{vs} = \frac{V_L}{V_s} = 44.62$ Ans

Q. For the given 2-stage RC-coupled amplifier circuit, find the overall voltage gain. (Assume $\beta_1 = \beta_2 = 100$)

Sol. The given circuit is:



The Thevenin's equivalent circuit for stage 1 is:



$R_{th} = \frac{12 \times 56}{12+56}$

$\therefore R_{th} = 9.88\ K$

$V_{th} = V_{cc} \times \frac{12}{56+12}$

$= 2.647V$

$\therefore$ Applying KVL in input loop, we get;

$V_{th} = R_{th} I_B + V_{BE} + I_E R_E$

$\Rightarrow 2.647 = 9.88 \times \frac{I_E}{100} + 0.7 + I_E$

$\Rightarrow 2.647 - 0.7 = 1.0988 I_E$

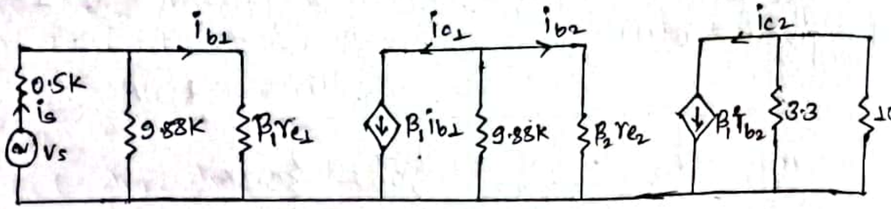
$\Rightarrow \frac{1.947}{1.0988} = I_E$

$\therefore I_E = 1.77mA$

Now,

$r_e = \frac{26mV}{1.77mA} = 14.69\Omega = 0.01469k\Omega$

The re-model of given RC coupled amplifier is:-



Now,

$R_{b2} = \beta_2 (r_{e2}) = 100 \times 0.01469 = 1.469 k\Omega$

$R_{b1} = \beta_1 r_{e1} = 100 \times 0.01469 = 1.469 k\Omega$

$r_{in1} = 0.5 + (9.88 // R_{b1}) = 0.5 + \left(\frac{9.88 \times 1.469}{9.88 + 1.469}\right)$

$= 1.778 k\Omega$

We know,

$V_L = -i_{c2}(3.3 // 10)$

$\Rightarrow V_L = -i_{c2}\left(\frac{3.3 \times 10}{3.3 + 10}\right)$

$\Rightarrow V_L = -2.48\, i_{c2}$

$\Rightarrow V_L = -2.48 \times \beta i_{b2}$

$\Rightarrow V_L = -248\, i_{b2}$

$\Rightarrow V_L = -248 \times (-i_{c1}) \times \frac{9.88}{9.88 + R_{b2}}$

$\Rightarrow V_L = 248 \times \frac{9.88}{9.88 + 1.469} \times i_{c1}$

$\Rightarrow V_L = 215.89 \times \beta_1 i_{b1}$

$\Rightarrow V_L = 215.89 \times 100 \times i_{b1}$

$\Rightarrow V_L = 21589\, i_{b1}$

$\Rightarrow V_L = 21589 \times i_s \times \frac{9.88}{9.88 + R_{b1}}$

$\Rightarrow V_L = 21589 \times \frac{9.88}{9.88 + 1.469} \times i_s$

$\Rightarrow V_L = 18794 \times \frac{V_s}{R_{in1}}$

$\Rightarrow V_L = \frac{18794}{1.778} \times V_s$

$\Rightarrow \frac{V_L}{V_s} = 10570$ Ans

Q. A 3-stage amplifier has a 1st stage voltage gain of 100, 2nd stage voltage gain of 200 and 3rd stage voltage gain ~~in dB~~ 400. Find total voltage gain in dB.

Sol, Given,

1st stage voltage gain, $A_{v1} = 100$

2nd stage voltage gain, $A_{v2} = 200$

3rd stage voltage gain, $A_{v3} = 400$

Now, The total voltage gain is $A_v = A_{v1} \times A_{v2} \times A_{v3}$

$= 100 \times 200 \times 400$

$\therefore A_v = 8000000$ Ans

Again,

Total voltage gain in dB, $A = 20 \log (A_v)$

$= 20 \log (8 \times 10^6)$

$\therefore A_{v_{dB}} = 138$ dB Ans

Q2. A multistage amplifier employs 5 stages each of which has a power gain of 30. What is the total gain of amplier in dB?

Sol Given:-

Power gain of each stage, $A_v = 30$

No. of stages, $N = 5$

$\therefore$ Total gain in dB $= N \times 20 \log (A_v)$

$= 5 \times 20 \log (30)$

$= 29.54 \times 5$

$= 147.71$ dB Ans

Pre-Board :

Q. A 3-stage amplifier has voltage gain of 50, 75 and 100 for the first, second and third Stage. Find overall gain in dB.

Soln Given:

Let A_{V1}, A_{V2} and A_{V3} be the voltage gain of 50, 75 and 100 respectively.

$\therefore$ Total voltage gain, $A_V = A_{V1} \times A_{V2} \times A_{V3}$

$= 50 \times 75 \times 100$

$= 375000$

$\therefore$ Total voltage gain, $A_V(dB) = 20\log(A_V)$

$= 20\log(375000)$

$= 111.48\,dB$ Ans

Feild Effect Transistor (Prabin Kushmi)

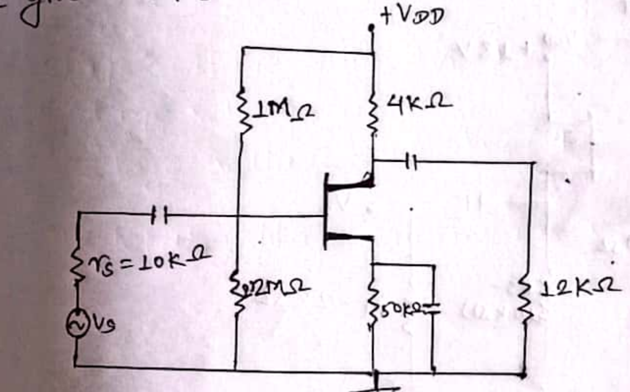
1. In JFET shown in figure, has transconductance $4000\mu S$ at its bias point. Its drain resistance is $100K\Omega$. Assuming small signal model, find the ovel all voltage gain,

Soln Given:

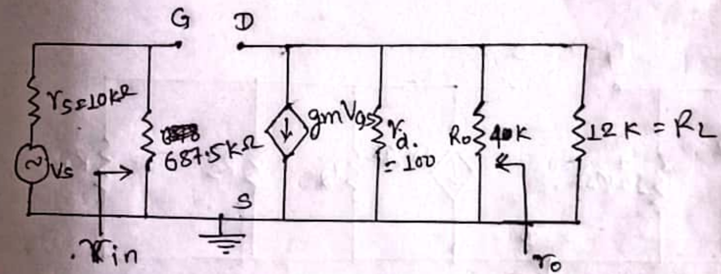
transconductance, $g_m = 4000\mu S = 4mS$

drain resistance, $r_d = 100K\Omega$

The given JFET is:



The small signal model of JFET is :



Here,

$r_{in} = 687.5K\Omega$

$r_o = r_d // R_D = \frac{100 \times 4}{100+4} = 3.846\,K\Omega$

$A_V = -g_m r_o = -4 \times 10^{-3} \times 3.846 \times 1000 = -15.384$ Ans

$A_{VS} = \frac{V_L}{V_S} = A_V \times \left(\frac{r_{in}}{r_s + r_{in}}\right) \times \left(\frac{R_L}{R_L + r_o}\right)$

$= -15.384 \times \left(\frac{687.5}{10+687.5}\right) \times \left(\frac{12}{12+3.846}\right) = -11.48$ Ans

2. Calculate Z_{in}, Z_o, A_v for the common drain JFET shown in figure. Given $I_{DSS} = 10mA$, $V_p = -5$ V, $r_d = 100K\Omega$.

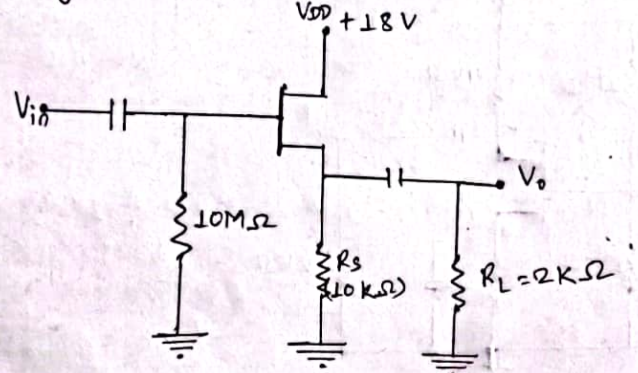
Sol^n Given:-

$I_{DSS} = 10mA = 10^{-2} A$

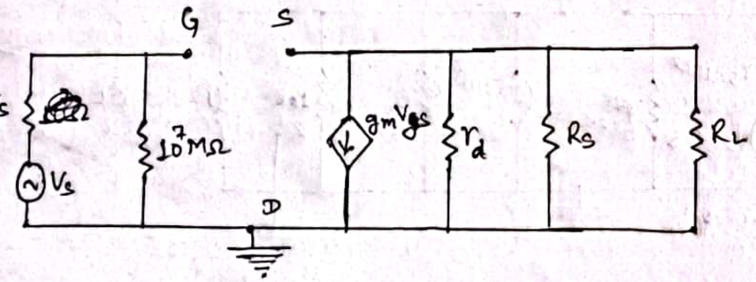
$V_p = -5V$

$r_d = 100 K\Omega$

The given JFET is:



The small signal model of given figure is



$$g_m = -\frac{2 I_{DSS}}{V_p}\left(1 - \frac{V_{GS}}{V_p}\right)$$

$\therefore V_{GS} = |V_p| - 0.7 = 5 - 0.7 = 4.3$

$\therefore V_{GS} = -4.3$

Now, $g_m = \frac{-2\times10^{-2}}{-5}\left(1 - \frac{-4.3}{-5}\right) = 4\times10^{-3}\times 0.14 = 0.56$ mS

$r_o = R_s // r_d + \frac{1}{g_m} = \frac{10\times100}{10+100} + \frac{1}{0.56mS}$

$\therefore r_o = 10.876 k\Omega$

Now, Voltage gain, $A_v = \frac{V_o}{V_{in}} = \frac{g_m V_{gs}(r_d//R_s)}{V_{gs} + g_m V_{gs}(R_s//r_d)}$

$= \frac{g_m (r_d//R_s)}{1 + g_m(R_s//r_d)}$

$= \frac{0.56\times(20//100)}{1+0.56\times(20//100)}$

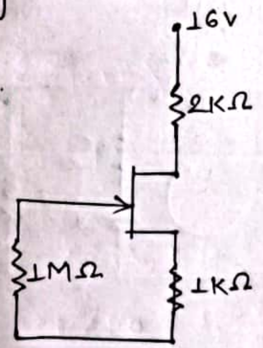
$= \frac{0.56\times16.67}{1+0.56\times16.67}$

$\therefore A_v = 0.9$ Ans

2015 - Fall

4.b. Determine I_{DQ} and V_{DSQ} for the given below. Given $I_{DSS} = 8mA$ and $V_p = -8V$.

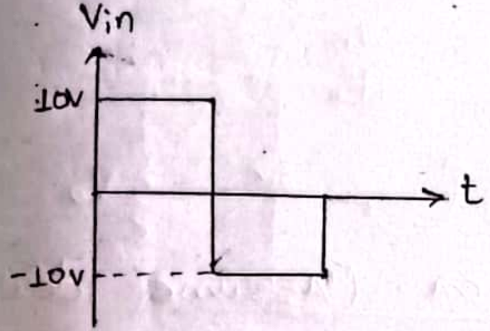
Sol^n The given circuit is:-

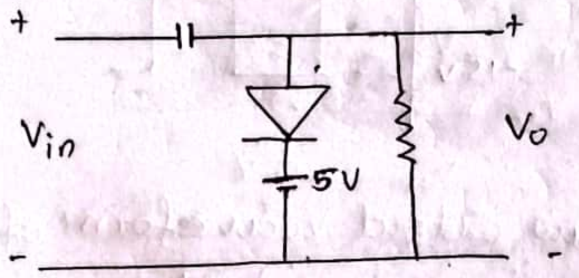


Clamper Circuit:

1. Draw output waveform of given circuit. (Assume ideal diode).

Sol: The given circuit is:





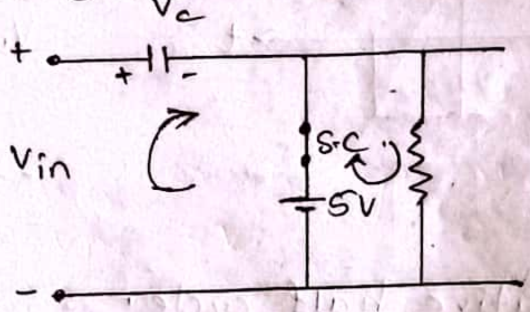
~~Here~~

i) For positive half cycle
⇒ Replace the diode by short circuit.

Applying KVL in input loop,
we get:- $V_{in} - V_c - 5 = 0$

$\Rightarrow 10 - 5 = V_c$

$\therefore V_c = 5V$



Applying KVL in output loop, we get:-

$5 - V_o = 0$

$V_o = 5V$

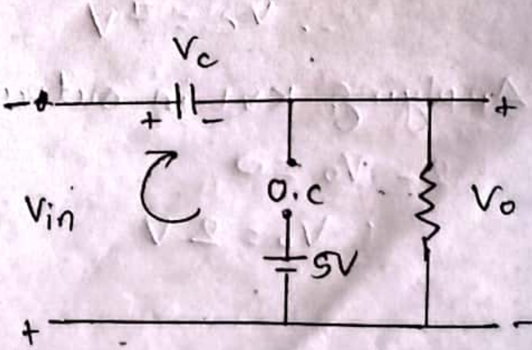
ii) For negative half cycle.
⇒ Replace the diode by open circuit.
Applying KVL in loop, we get;

$-V_{in} - V_c - V_o = 0$

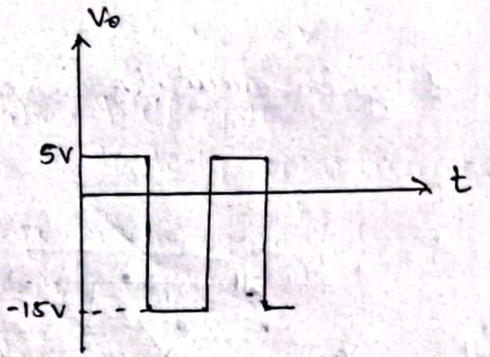
$\Rightarrow -V_o = V_{in} + V_c$

$\Rightarrow -V_o = 10 + 5$

$\therefore V_o = -15V$

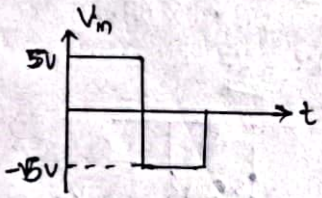


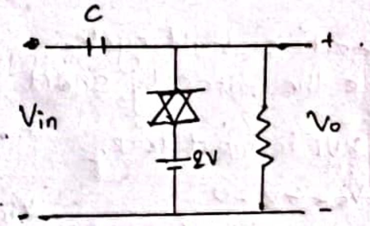
The output waveform is:



2. Draw output waveform of given. (Assume ideal diode.)

Sol The given circuit is:

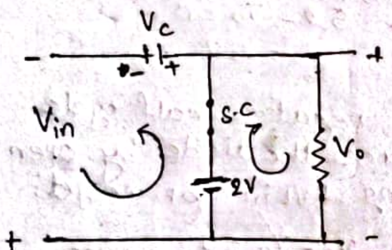




i) For positive half cycle.

⇒ Replace the diode by short circuit.

Applying KVL in input loop,



$V_{in} + \cancel{\ } 2 - V_c = 0$

⇒ $V_{in} + 2 - V_c = 0$

⇒ $5 + 2 - V_c = 0$

∴ $V_c = 7V$

Applying KVL in output loop,

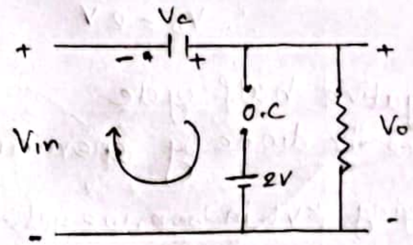
$2 - V_o = 0$

∴ $V_o = 2\ V$

ii) For negative half cycle

⇒ Replace the diode by open circuit.

Applying KVL in loop, we get;



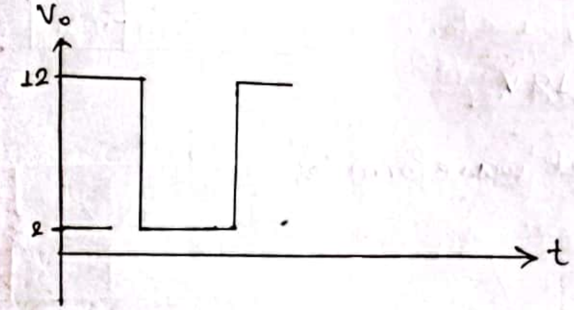
$V_{in} + V_c - V_o = 0$

⇒ $V_{in} + V_c = V_o$

⇒ $V_o = 5 + 7$

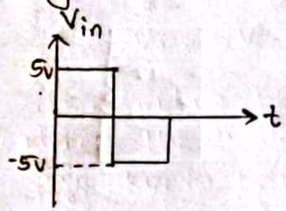
∴ $V_o = 12V$

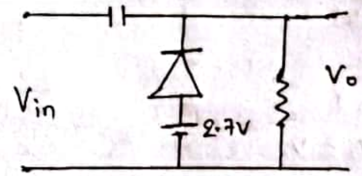
The output waveform is:



3. Draw the output waveform of given circuit. (Assume Si ~~ideal~~ diode)

Sol The given circuit is:

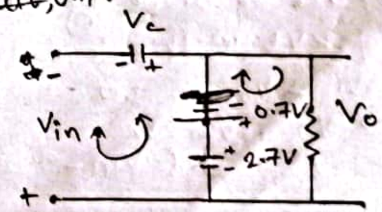




i) For positive half cycle.

⇒ Replace the diode by ~~short circuit~~, 0.7V (si diode).

Applying KVL in input loop, we get:



$V_{in} + 2.7 - 0.7 - V_c = 0$

⇒ $5 + 2 = V_c$

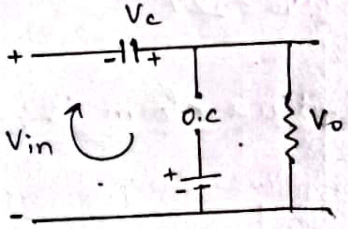
∴ $V_c = 7\ V$

Applying KVL in output loop, we get:-

$$2.7 - 0.7 - V_o = 0$$
$$\therefore V_o = 2V$$

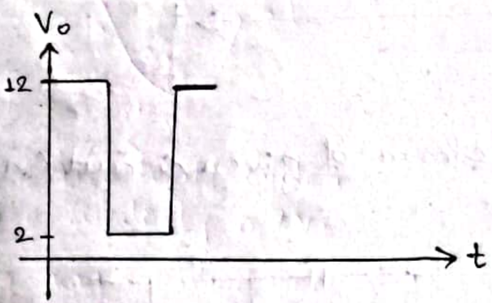
ii) For negative half cycle
→ Replace the diode by open circuit.

Applying KVL in loop, we get:-



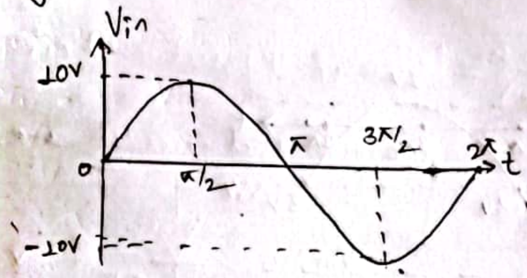
$$V_{in} + V_c - V_o = 0$$
$$\Rightarrow \cancel{V_{in} + 7 - 2 = 0}$$
$$\therefore \cancel{V_{in} = 5V}$$
$$\Rightarrow 5 + 7 = V_o$$
$$\therefore V_o = 12V$$

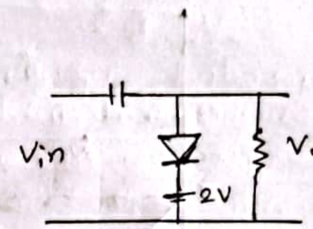
The given output waveform is:



4. Draw output waveform. (Assume ideal diode).

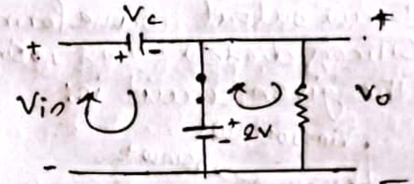
Sol: The given circuit is:-





i) For positive Half cycle of input:-
→ Replace the diode by short circuit

Apply KVL in input loop, we get;



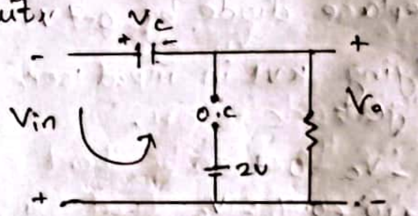
$$V_{in} - V_c - 2 = 0$$
$$\Rightarrow V_c = V_{in} - 2$$
$$\Rightarrow V_c = 10 - 2 = 8$$

Apply KVL in output loop, we get;

$$2 - V_o = 0$$
$$\therefore V_o = 2$$

ii) For negative half cycle of input.
→ Replace the diode by open circuit.

Apply KVL in the loop, we get:-

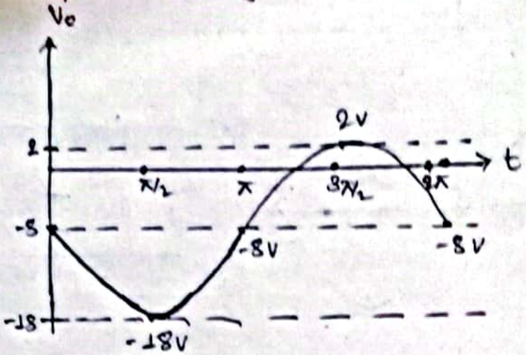


$$V_{in} + V_c + V_o = 0$$
$$\Rightarrow V_{in} + 8 + V_o = 0$$
$$\therefore V_o = -V_{in} - 8$$

Now

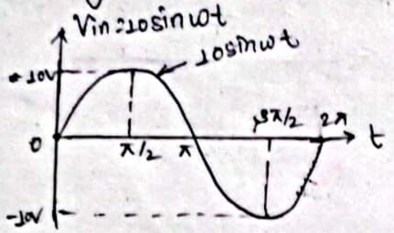
ωt	0	$\pi/2$	π	$3\pi/2$	2π
V_{in}	0	10	0	-10	0
V_o	-8	-18	-8	2	-8

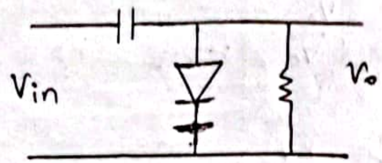
Thus the output waveform will be.



5. Draw output waveform of given circuit. (Assume Silicon diode), $V_{in} = 10\sin\omega t$.

Sol The given circuit is:

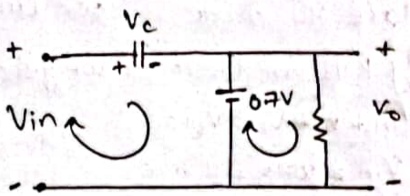




i) For positive half cycle:

⇒ Replace diode by 0.7V.

Applying KVL in input loop, we get;



$V_{in} - V_c - 0.7 = 0$

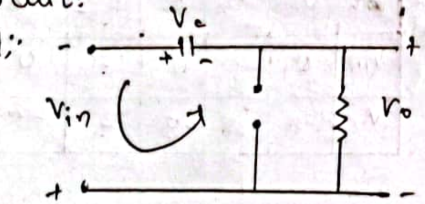
⇒ $+10 - 0.7 = V_c$

$\therefore V_c = 9.3V$

ii) For negative half cycle:

⇒ Replace the diode by Open circuit:

Apply KVL in the loop, we get;



$V_{in} + V_o + V_c = 0$

⇒ $V_o = -V_{in} - V_c$

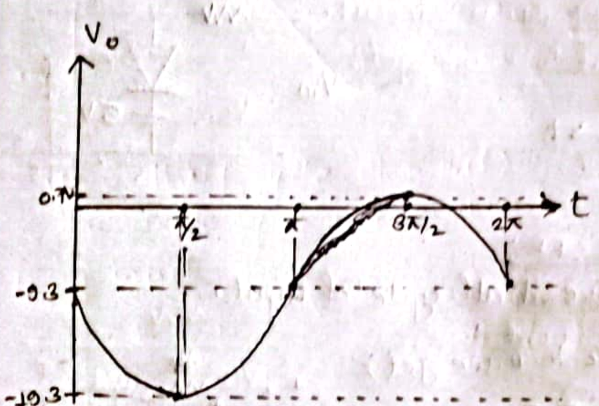
⇒ $V_o = (-V_{in} - 9.3)V$

wt	0	$\pi/2$	π	$3\pi/2$	2π
Vin	0	10	0	-10	0
V_o	-9.3	19.3	-9.3	0.7	-9.3

Note

$\boxed{V_o = -V_{in} - 9.3}$

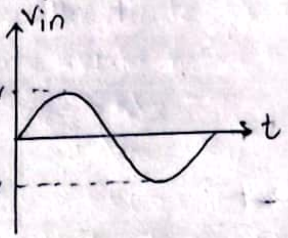
The output waveform will be

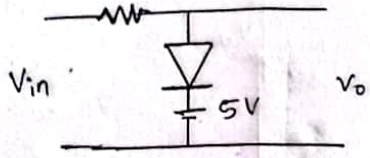


~~Clamp~~ Clipping Circuits (Prabin chaudhary)

1. Draw output waveform for following ~~ip~~ input and circuit. (Assume ideal diode)

Sol The given input ~~and~~ waveform and circuit are:





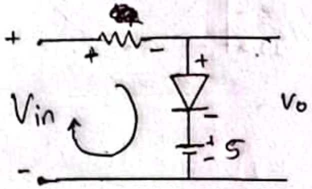
consider a positive half cycle of input.

Applying KVL in the loop, we get:

$V_{in} - V_D - 5 = 0$

$\Rightarrow V_{in} - 5 = V_D$

$\therefore V_D = V_{in} - 5$



Condition for diode to be ON.

$V_D > 0$

$\Rightarrow V_{in} - 5 > 0$

$\therefore V_{in} > 5V$

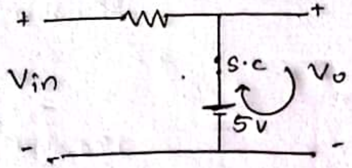
i) If the diode is ON.

$\Rightarrow$ Replace the diode by short circuit.

Applying KVL in output loop,

$5 - V_o = 0$

$\therefore V_o = 5V$



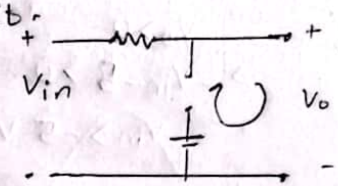
ii) If the diode is OFF,

$\Rightarrow$ Replace the diode by open circuit.

Applying KVL in output loop,

$V_{in} - V_o = 0$

$\therefore V_o = V_{in}$



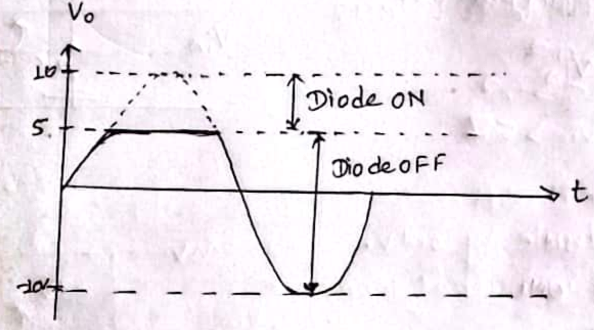
Now,

condition is, $V_{in} > 5$

Diode ON, $V_o = 5V$

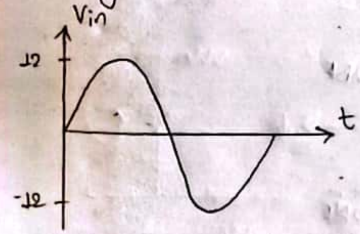
Diode OFF, $V_o = V_{in}$

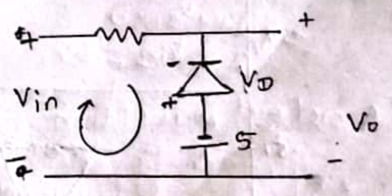
Thus the output waveform will be



2. Draw output waveform of given input waveform and circuit. (Assume ideal diode)

Sol The given waveform and circuit is;





Applying KVL in input loop, we get:-

$V_{in} + V_D + 5 = 0$

$\Rightarrow V_D = -V_{in} - 5$

Condition for diode to be ON,

$V_D > 0$

$\Rightarrow -V_{in} - \cancel{5} > 0$

$\Rightarrow V_{in} > -5\,V \Rightarrow \cancel{V_{in} < 5V}$

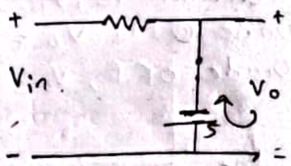
<u>If the diode is ON:</u>

⇒ Replace the diode by open circuit.

Applying KVL in input loop, we get:

$-5 + V_o = 0$

$\therefore V_o = -5V$



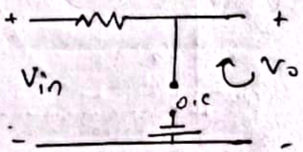
<u>If the diode is OFF:</u>

⇒ Replace the diode by short circuit.

Applying KVL in output loop,

$V_{in} - V_o = 0$

$\therefore V_o = V_{in}$



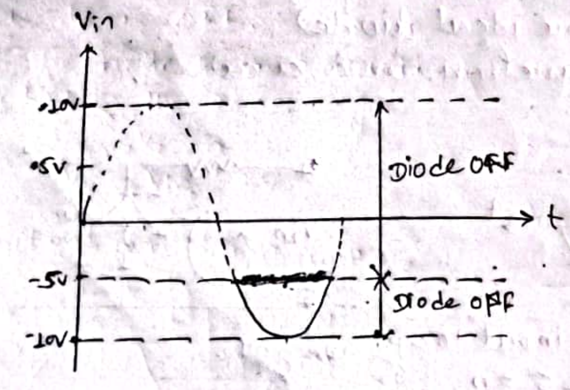
Condition is,

$V_{in} < 5V$

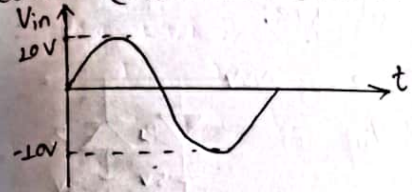
$V_o = -5V$ for ON

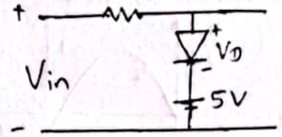
$V_o = V_{in}$ for ON

The output waveform will be



3. Draw the output waveform of given input waveform and circuit. (Assume silicon diode)



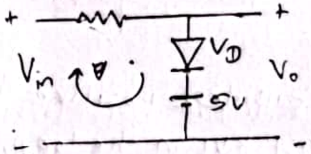


⇒ Consider positive half cycle of input.

Applying KVL in input loop, we get:

$V_{in} - V_D - 5 = 0$

$\therefore V_D = V_{in} - 5$



Condition for diode to be ON,

$V_D > 0.7$

$\Rightarrow V_{in} - 5 > 0.7$

$\therefore V_{in} > 5.7\,V$

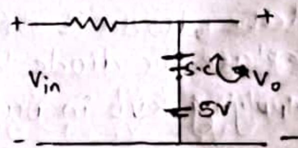
→ If the diode is ON:

⇒ Replace the diode by short circuit.

Applying KVL in output loop,

$5 - V_o + 0.7$

$\therefore V_o = 5 + 0.7 = 5.7V$



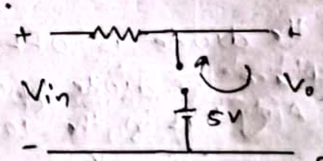
→ If the diode is OFF.

⇒ Replace the diode by open circuit.

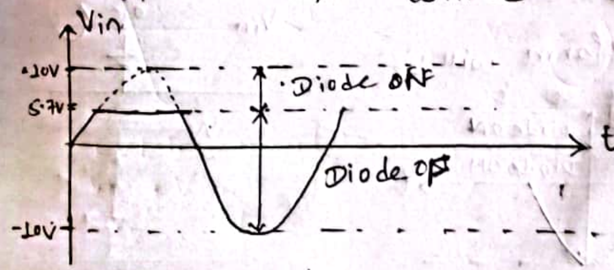
Applying KVL in output loop, we get:

$V_o = V_{in}$

$\therefore V_o = V_{in}$

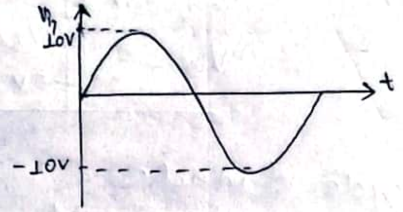


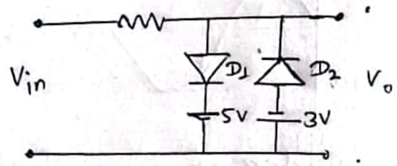
Thus the output waveform will be



4. Draw output waveform. (Assume silicon diode)

⇒ The input waveform and circuit are;



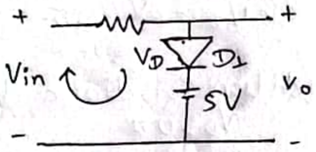


~~Condi~~ Consider positive half cycle of diode D_1 only.

Applying KVL in the loop, we get;

$V_{in} - V_D - 5V = 0$

$\therefore V_D = V_{in} - 5V$



Condition for diode to be ON,

$V_D > 0.7$

$\Rightarrow V_{in} - 5 > 0.7$

$\therefore V_{in} > 5.7$

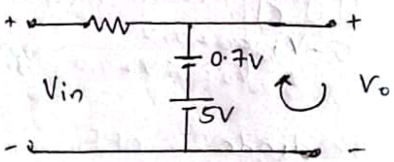
i) If the diode is ON

⇒ Replace the diode by 0.7V.

Applying KVL in Output loop,

$5 + 0.7 - V_o = 0$

$\therefore V_o = 5.7V$



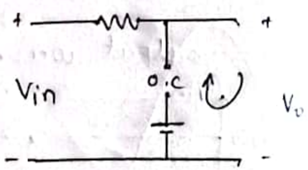
ii) If the diode is OFF.

⇒ Replace the diode by open circuit.

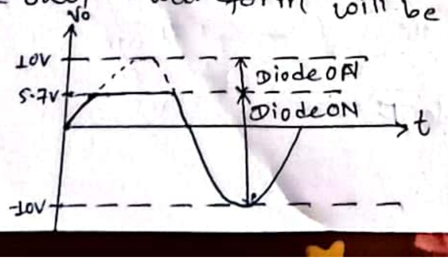
Apply KVL in the loop, we get;

$V_{in} - V_o = 0$

$\therefore V_o = V_{in}$



Thus the output waveform will be

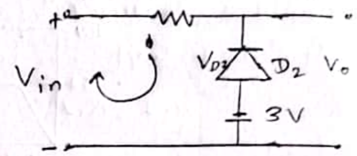


Again Consider ~~negative~~ positive half cycle of diode D_2 only.

Apply KVL in the loop, we get:

$V_{in} + V_{D2} + 3 = 0$

$\rightarrow V_D = -V_{in} - 3$



Condition for diode to be ON,

$V_D > 0.7$

$\Rightarrow -V_{in} - 3 > 0.7$

$\therefore V_{in} > -3.7$

$\therefore$ ~~$V_{in} < -3.7V$~~

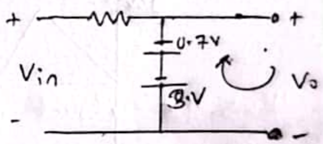
If diode is ON:

Replace the diode by 0.7V.

Applying KVL in output loop, we get:

$-3 - 0.7 - V_o = 0$

$\therefore V_o = -3.7V$



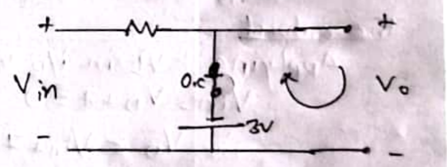
If diode is OFF:

Replace the diode by open circuit.

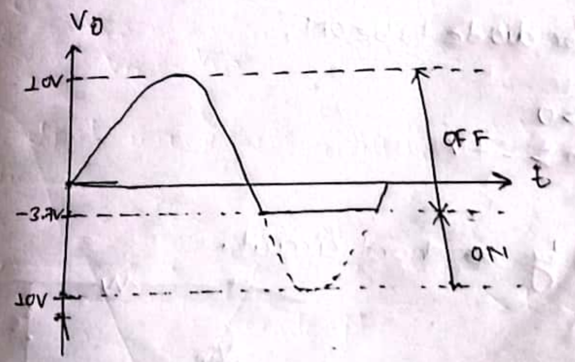
Applying KVL in the output loop,

$V_{in} - V_o = 0$

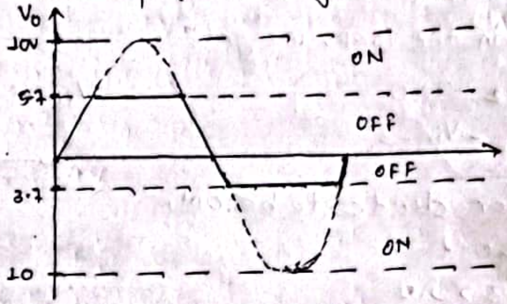
$\therefore V_o = V_{in}$



So the output waveform for D_2 will be

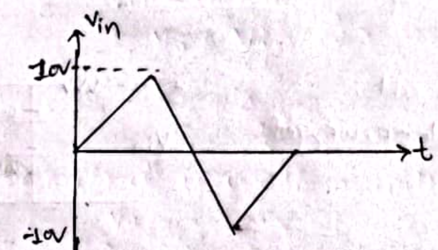


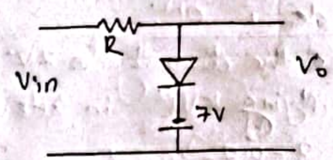
Thus the final out put waveform will be



5. Draw output waveform. (assume ideal diode).

Sol





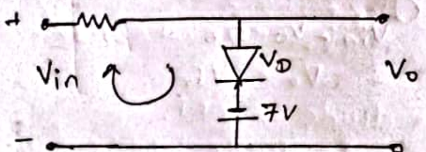
Consider positive half cycle of diode only

~~The output~~

Applying KVL in i/p loop,

$V_{in} - V_D + 7 = 0$

$\therefore V_D = V_{in} + 7$



~~If the diode is ON~~;

Condition for diode to be ON,

$V_D > 0$

$\Rightarrow V_{in} + 7 > 0$

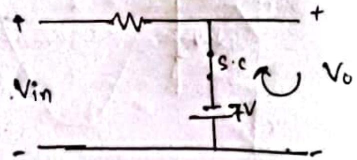
$\therefore V_{in} > -7V$

If the diode is ON

⇒ Replace the diode by short circuit.

$-7 - V_0 = 0$

$\therefore V_0 = -7V$



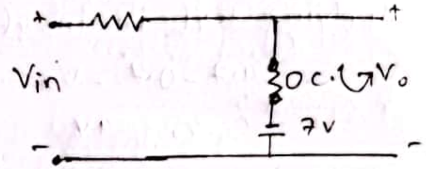
If the diode is OFF

Replace the diode by open circuit.

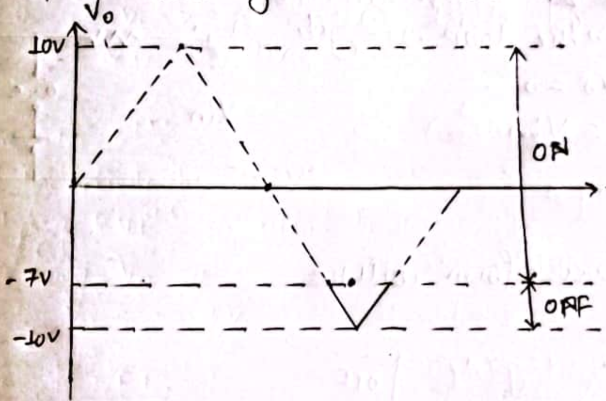
Applying KVL in loop,

$V_{in} - V_0 = 0$

$\therefore V_0 = V_{in}$

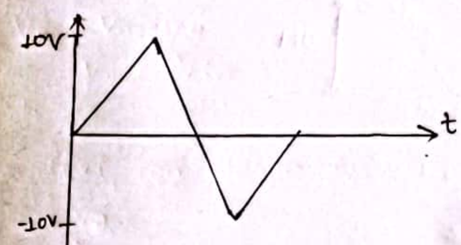


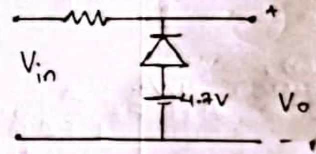
The output waveform will be.



6. Draw output waveform. (Assume practical diode)

⇒



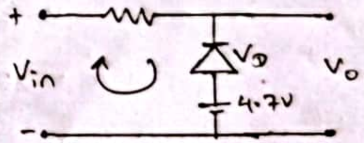


Sol Consider for positive half cycle of a diode.

Applying KVL in the loop, we get;

$V_{in} + V_D - 4.7\ V = 0$

$\therefore V_D = -V_{in} + 4.7$



Condition for diode to be ON,

$V_D > 0.7$

$\Rightarrow -V_{in} + 4.7 > 0.7$

$\therefore V_{in} < 4\ V$

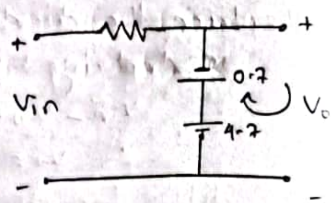
If the diode is ON

→ Replace the diode by 0.7V.

Applying KVL in input loop, we get:

$4.7 - 0.7 - V_o = 0$

$\therefore V_o = 4V$



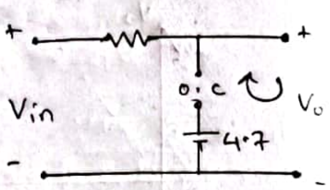
If the diode is OFF:

→ Replace the diode by open circuit.

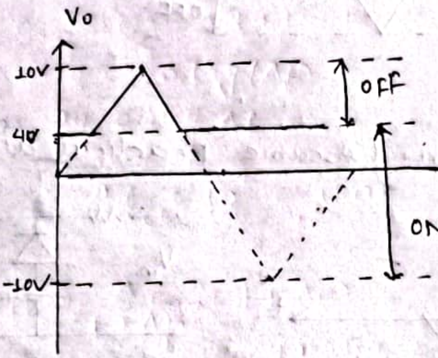
Applying KVL in the loop, we get,

$V_{in} - V_o = 0$

$\therefore V_o = V_{in}$



Thus the output waveform will be



Feedback Amplifier (Prabin)

1. The voltage gain of an amplifier without feedback is 3000. Calculate the voltage gain of the amplifier if negative feedback is introduced in the circuit. (Given feedback factor = 0.01)

Sol^n Given;

~~without~~ voltage gain, $A = 3000$

feedback factor, $\beta = 0.01$

voltage gain with feedback, $A_f = ?$

we know that,

$$A_f = \frac{A}{1+A\beta}$$

$$= \frac{3000}{1+3000\times0.01}$$

$$\therefore A_f = 96.77 \quad \text{Ans}$$

2. The overall gain of an multistage amplifier is 140. When negative feedback is applied the gain is reduced to 17.5. Find the fraction of the output that is feedback to the input.

Sol^n Given;

gain without feedback, $A = 140$

gain with feedback, $A_f = 17.5$

feedback ratio, $\beta = ?$

we know that,

$$A_f = \frac{A}{1+A\beta}$$

$$\Rightarrow 17.5 = \frac{140}{1+140\beta}$$

$$\Rightarrow 1+140\beta = \frac{140}{17.5}$$

$$\Rightarrow 140\beta = 8-1 = 7$$

$$\therefore \beta = \frac{7}{140} = \frac{1}{20} = 0.05 \quad \text{Ans}$$

3. The voltage gain of an amplifier without feedback is 60dB. It decreases to 40 dB with feed-back. Calculate the % of output which is feed back to to the input.

Sol Given:-

~~feedbackwithout~~

In dB, voltage gain without feedback, $A = 60dB$

voltage gain with feedback, $A_f = 40dB$

feedback ratio, $\beta = ?$

we know,

$$60 = 20\log(A)$$

$$\Rightarrow \frac{60}{20} = \log A$$

$$\therefore A = 10^3 = 1000$$

Also,

$$40 = 20\log(A_f)$$

$$\Rightarrow \frac{40}{20} = \log A_f$$

$$\therefore A_f = 10^2 = 1000$$

Now,

$$A_f = \frac{A}{1+A\beta}$$

$$\Rightarrow 100 = \frac{1000}{1+1000\beta}$$

$$\Rightarrow 1+1000\beta = \frac{1000}{100}$$

$$\Rightarrow 1000\beta = 10-1$$

$$\Rightarrow \beta = \frac{9}{1000}$$

$$\therefore \beta = 0.009$$ Ans

4. A negative feedback of 0.2% is applied to an amp gain of voltage ~~60f~~ 60dB. Calculate the % change in the overall gain of feedback amplifier if the internal amplifier is subjected to a gain of the reduction of 15%.

Sol Given:

~~$\frac{dA_f}{A_f} = 0.2\%$~~

~~$A_f = 60dB$~~

feed back ratio %, $\beta = 0.2\% = 0.002$

voltage gain in dB with feedback, $A = 60dB$

$$\frac{dA}{A} = 15\%$$

Now,

$$60 = 20\log(A)$$

$$\Rightarrow \frac{60}{20} = \log A$$

$$\therefore A = 1000$$

Now,

$$\frac{dA_f}{A_f} = \frac{dA}{A} \times \frac{1}{1+A\beta}$$

$$\Rightarrow \frac{dA_f}{A_f} = \frac{15\%}{1+1000\times 0.002} = 5\%$$

$$\therefore \frac{dA_f}{A_f} = 5\%$$ Ans

5. An amplifier has the midband gain of 1500 and BW of 4MHz. The midband gain reduces to 150 when a negative feedback is applied. Determine the value of feedback & BW.

Soln Given :-

Midband gain, $A = 1500$

Bandwidth, $BW = 4MHz$

Midband gain with feedback, $A_f = 150$

we know,

$$A_f = \frac{A}{1+A\beta}$$

$$\Rightarrow \frac{A_f}{A} = \frac{1}{1+A\beta}$$

$$\Rightarrow \frac{150}{1500} = \frac{1}{1+1500\beta}$$

$$\Rightarrow 10 = 1+1500\beta$$

$$\Rightarrow \frac{10-1}{1500} = \beta$$

$$\therefore \beta = 0.006$$

$\therefore$ feedback ratio, $\beta = 0.006 = 0.6\%$

Again,

$$BW_f = BW(1+A\beta)$$

$$= 4\times10^6 (1+1500\times0.006)$$

$\therefore BW_f = 40MHz$ Ans